VIETNAM NATIONAL UNIVERSITY OF HOCHIMINH CITY
THE INTERNATIONAL UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BUILDING AN AUTOMATED SYSTEM FOR IDENTIFYING VIETNAMESE LEGAL DOCUMENTS IN EDUCATION USING AI AND LLM

By: Phan Anh Kiet
ID: MITIU23202
Advisor: Prof. Nguyen Van Sinh

A thesis submitted to the School of Computer Science and Engineering in partial fulfillment of the requirements for the degree of Master of Information Technology

Ho Chi Minh City, Vietnam
2025

BUILDING AN AUTOMATED SYSTEM FOR IDENTIFYING VIETNAMESE LEGAL DOCUMENTS IN EDUCATION USING AI AND LLM

APPROVED BY:

_____

Nguyen Van Sinh, Assoc. Prof

_____

(*Typed Committee name here*)

_____

(*Typed Committee name here*)

_____

(*Typed Committee name here*)

_____

(*Typed Committee name here*)

THESIS COMMITTEE
(Whichever applies)

ACKNOWLEGMENTS

I extend my sincere gratitude and appreciation to Associate Professor Nguyen Van Sinh for providing me with the invaluable opportunity to undertake this thesis. His guidance and support have been instrumental in the success of this work. I am also thankful to all the lecturers at the School of Computer Science and Engineering at Vietnam National University HCMC—International University. Their instruction has significantly contributed to my academic development.

Special thanks are due to my family, whose unwavering love and support have been my greatest source of strength. They have always stood by my side, helping me maintain focus and motivation, especially during challenging times. I am profoundly thankful for all they have done for me.

This project not only represents my individual efforts but also stands as a collaborative achievement. I deeply appreciate the contributions of my professor, lecturers, and family, all of whom have played essential roles in bringing this work to fruition. My heartfelt thanks extend to each and every one of them.

This thesis is funded by Vietnam National University Ho Chi Minh City (VNU-HCM) under grant number DS2025-28-03. We would like to thank for the fund. This thesis is also supported by The Central Interdisciplinary Laboratory in Electronics and Information Technology (AI and Cooperation Robot), International University - VNU-HCM. We would like to thank for supporting the machines in experiments.

TABLE OF CONTENTS

ACKNOWLEGMENTS 3
TABLE OF CONTENTS 4
TABLE OF ABBREVIATIONS 6
LIST OF FIGURES 8
LIST OF TABLES 9
ABSTRACT 10
CHAPTER 1: INTRODUCTION 11
1.1. Background 11
1.2. Problem Statement 11
1.3. Scope and Objectives 12
1.4. Assumption and Solution 12
1.5. Structure of thesis 13
CHAPTER 2: LITURATURE REVIEW 14
2.1. Digital transformation 14
2.2. Standard structure of Vietnamese administrative documents 15
2.3. Page segmentation and image preprocessing for OCR 18
2.3.1. Page Segmentation 18
2.3.2. Image Preprocessing 19
2.3.3. Cross-Domain Applications of Image Processing 21
2.4. Natural language processing for legal documents 22
2.4.1. Training data requirements for legal NLP 22
2.4.2. Popular methods and models implement for training data 22
2.5. OCR technology and development 23
2.5.1. Overview of OCR 23
2.5.2. Practical assessment using OCR 25
2.5.3. Tesseract OCR for Vietnamese 26
2.5.4. Hand-writing text OCR for Vietnamese 27
2.6. Techniques for OCR recognition 29
2.6.1. Thresholding for robust binarization 29
2.6.2. Detecting circles using Hough Circle Transform 30
2.6.3. Find contours to separate text layout blocks 31
2.6.4. Classification and recognition using Support Vector Machines (SVM) 33
CHAPTER 3: METHODOLOGY 35
3.1. Overview 35
3.1.1. Proposed method 35
3.1.2. Platform Overview 36

3.2. Collect data 37
3.3. Image processing 39
3.4. Text Recognition and Extraction 42
3.4.1. Advanced Preprocessing for OCR 42
3.4.2. Text Region Classification Using SVM 43
3.4.3. Print-Text Recognition Using Tesseract OCR 44
3.4.4. Handwriting-Text Recognition Using CRNN 45
3.4.5. Structured Data into an Administrative Form 48
3.5. Convert into XML structure 49
3.5.1. Conversion objective 49
3.5.2. Labeling Components 50
3.5.3. Conversion Process 51
CHAPTER 4: IMPLEMENTATION AND RESULTS 53
4.1. Technology platform implement 53
4.2. System implement 55
4.3. Interface and Feature functions 57
CHAPTER 5: DISCUSSION AND EVALUATION 60
5.1. Evaluation Metrics 60
5.2. Quantitative results 60
5.3. Feature comparison 62
CHAPTER 6: CONCLUSION AND FUTURE WORK 63
6.1. Conclusion 63
6.2. Future work 64
REFERENCES 65

TABLE OF ABBREVIATIONS

No.	Abbreviation	Full Form
1	**AI**	Artificial Intelligence
2	**LLM**	Large Language Model
3	**GenAI**	Generative Artificial Intelligence
4	**OCR**	Optical Character Recognition
5	**NLP**	Natural Language Processing
6	**XML**	eXtensible Markup Language
7	**MOET**	Ministry of Education and Training
8	**PDF**	Portable Document Format
9	**ETL**	Extract, Transform, Load
10	**API**	Application Programming Interface
11	**SVM**	Support Vector Machine
12	**CRNN**	Convolutional Recurrent Neural Network
13	**CNN**	Convolutional Neural Network
14	**CTC**	Connectionist Temporal Classification
15	**LSTM**	Long Short-Term Memory
16	**HTML**	HyperText Markup Language
17	**CSS**	Cascading Style Sheets
18	**JSON**	JavaScript Object Notation
19	**CSV**	Comma-Separated Values
20	**RNN**	Recurrent Neural Network
21	**DL**	Deep Learning
22	**ML**	Machine Learning
23	**PyTorch**	Python Torch (Deep Learning Framework)
24	**OpenCV**	Open Source Computer Vision Library
25	**TF**	TensorFlow

LIST OF FIGURES

Figure 1: Structure of thesis 13
Figure 2: Example of an administrative document. 16
Figure 3: Structure format of an administrative document. 17
Figure 4: Legal Document after Noise removal. 20
Figure 5: The development of OCR. 24
Figure 6: Block diagram of the overall architecture of Tesseract [20] 26
Figure 7: Proposed workflow for legal text processing and tagging 35
Figure 8: National Database of Legal Documents Sitemap 37
Figure 9: The original image of legal documents included many noise 39
Figure 10: The image after Denoising 40
Figure 11: Thresholding the image to separate the text in image 41
Figure 12: OCR processing for extracting raw text from pdf document file. 42
Figure 13: SVM to separate the handwriting-text and printed-text 44
Figure 14 Print Recognition Output Using Tesseract OCR 44
Figure 15: Processing for convert to XML 49
Figure 16: The web-app for implementing all steps of proposed method 56
Figure 17: Interface for Uploading pdf file 57
Figure 18: Extract the text from the legal document pdf file 58
Figure 19: XML data structure output 59

LIST OF TABLES

Table 1: Table of Digital transformation 15
Table 2: Table of Page segmentation method 18
Table 3: Table of Hough Circle process 30
Table 4: Table of Contours to separate layout blocks 32
Table 5: Table of Handwriting-Text Recognition CRNN layer 47
Table 6: XML tags for structured legal document components 51
Table 7: Measure the error ratio of words and characters performed 61
Table 8: Comparison between applications and existing applications in practice. 62

ABSTRACT

Digital transformation in education requires an intelligent system to extract, standardize and analyze large volumes of legal documents in many formats. In Vietnam, legal documents in the education have complex structures, including national emblems, issuing agencies, symbols, dates of issuance, clauses, appendices and sometimes signatures or handwritten notes. Processing these documents poses many challenges in information retrieval and legal data management. This thesis proposes an automated system integrating Natural Language Processing (NLP), Optical Character Recognition (OCR) and Image Processing to extract and label the structure of Vietnamese legal documents in education. The system performs:(1) Collecting PDF legal documents from the portal of the Ministry of Education and Training (MOET);(2) Labeling data according to categories: document type, issuing agency, article number, main content;(3) structuring the extracted text into XML format. This system lays the foundation for an AI-based legal lookup platform, contributing to promoting the goal of digital transformation of Vietnam's education sector.

CHAPTER 1: INTRODUCTION

1.1. Background

In the context of digital transformation in the education sector, the need of information from legal documents is becoming increasingly important. Legal documents of the Ministry of Education and Training (MOET) often exist in many different formats such as PDF, scanned images, or digitized text from printed documents. The volume of this data is increasing, requiring an intelligent processing system to standardize and store information effectively.

Natural Language Processing (NLP) technology combined with Optical Character Recognition (OCR) and image processing allows the extraction and conversion of text data from unstructured to structured formats. This not only helps to increase processing speed but also improves the accuracy of information recognition. In addition, the rapid development of Large Language Models (LLMs) and Generative AI opens up the possibility of building intelligent search and decision-making systems. When combined with standardized legal databases, these models can support contextual analysis, question answering, text summarization, and legal suggestions in the education field. The system proposed in this study aims to create an optimal data platform for smart system, contributing to promoting the process of legal search quickly and accurately.

1.2. Problem Statement

The system of legal documents in the field of education in Vietnam is very large, complex and regularly updated. This complexity poses significant challenges in researching, processing and extracting information from these documents. Therefore, the system of modern technology is very important to improve the efficiency of management, searching and analyzing legal content:

- Data formats: Most documents are stored as scanned PDFs or image formats, making it difficult to search for specific content.
- Lack of data standardization: Even digitized data often suffers from inconsistent separation, labeling, and storage, making it difficult to integrate into intelligent search systems.
- Inconsistent OCR accuracy: Documents are poorly scanned, contain handwritten text, or contain special symbols that are difficult to recognize.

- Limited search and semantic analysis capabilities: Current systems primarily support basic keyword searches and do not leverage contextual analysis or provide feedback based on text content.

The main objective is to develop a comprehensive system that can automatically collect, process, standardize, and analyze data from legal documents. The system will provide the foundational support needed to enhance a Large Language Model (LLM) that leverages insights derived from the processed data, facilitating more advanced applications in the realm of legal analysis.

1.3. Scope and Objectives

The system aimed at the automatic recognition and standardization of legal documents within the education sector, specifically focusing on those issued by the Ministry of Education and Training (MOET). The system is designed to handle both electronic documents and to recognize and extract information from PDF files and scanned images. The processing procedure comprises several key stages: document collection, pre-processing, character recognition (OCR), structural analysis, and the storage of data in structured formats such as XML or JSON. The system involves three primary tasks:

1. Collecting legal document data from official sources.
2. Identifying and extracting content from the documents using OCR technology.
3. Converting the extracted data into a structured XML format.

1.4. Assumption and Solution

Document digitization projects involve many complex issues, from image scanning, character recognition, data processing to document storage and management on the system. To reduce the development workload and optimize implementation time, a reasonable approach is to utilize and combine existing libraries and frameworks related to the project.

Using developed and proven tools and platforms helps:

- Take advantage of optimized technology and community support.
- Increase stability and security thanks to widely tested solutions.
- Easily expand and upgrade in the future when there is a need to integrate new features.

By properly selecting and integrating popular open source libraries and frameworks, the system can quickly meet the requirements of document scanning, recognition, storage and searching, while ensuring the efficiency and quality of the digitization process.

1.5. Structure of thesis

This entire research paper is divided into 6 chapters:

Chapter 1, "Introduction". This chapter will mainly go through the brief introduction and general information about the research topic.

Chapter 2, "Literature Review". This chapter includes the content of research works related to topic and previous research parts that can be evaluated, used and inherited for the research topic.

Chapter 3, "Methodology" .This chapter will talk about the research method and the content of Building an Automated System for Identifying Vietnamese Legal Documents in Education using AI and LLM.

Chapter 4, "Implementation and Results" .This chapter describes the implementation of system, the necessary libraries and tools, then the research results obtained for the above system.

Chapter 5, " Discussion and Evaluation". This chapter analyzes in more depth the results obtained, based on the results, a specific assessment will be made for that result.

Chapter 6, "Conclusion and Future Work". This chapter summarizes the entire research and then gives directions for future research.

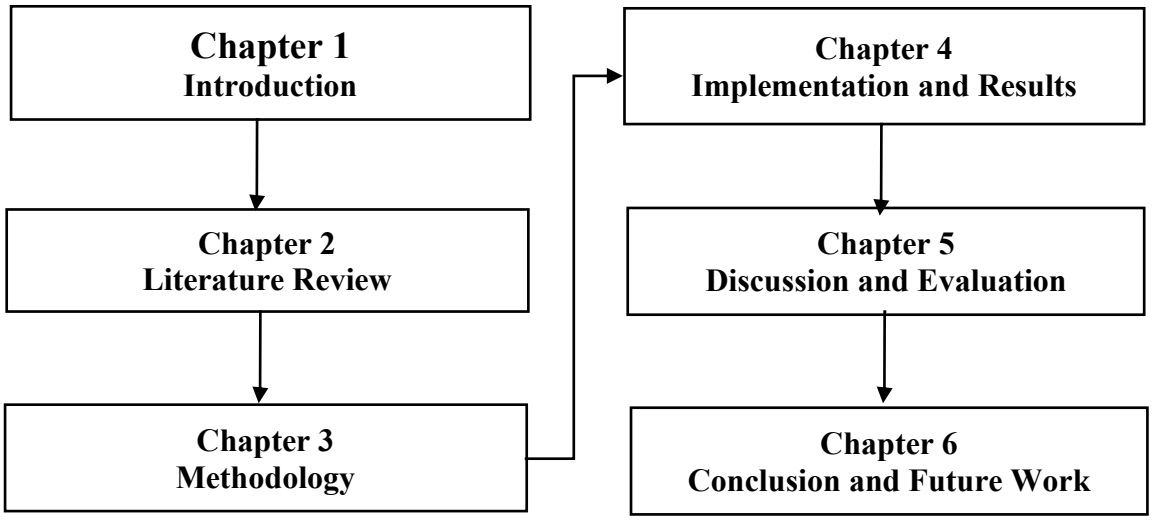


Figure 1: Structure of thesis

CHAPTER 2: LITURATURE REVIEW

2.1. Digital transformation

In the context of globalization and the strong development of digital technology, digital transformation has become an inevitable trend in most fields, especially administrative document management and education. According to [1], the digitalization of administrative documents not only helps optimize the management process but is also an important step in administrative reform and modernization of the public administration system. At the strategic level, Siebel [2] emphasized that digital transformation is not simply the application of technology but the process of comprehensively restructuring the way an organization operates to adapt and develop in the "era of mass extinction" of traditional models. At the same time, research by Ziyadin et al. [3] shows that in the field of education, digital transformation plays a key role in improving the efficiency of information management, supporting decision making and improving the learning experience of learners.

In the educational environment, digital transformation applications have been strongly deployed, from training management to student learning outcomes assessment. Johan et al. [4] proposed a digitalization and grading system for graduation projects to ensure transparency and objectivity in assessment. Next, this group of authors [5] deployed a document storage solution on a cloud computing platform, allowing quick access, minimizing the risk of data loss and enhancing remote collaboration between lecturers and students.

In the entire document digitization process, optical character recognition (OCR) technology plays a particularly important role, especially in converting data from image format to searchable and editable text. The open source OCR tool Tesseract is chosen by many studies due to its flexibility, multilingual support and strong development community. Research by Pate and Patel [6] demonstrates that Tesseract can be effectively integrated into complete digitization systems, automating the process of extracting data from printed and scanned documents, thereby significantly reducing manual processing time and costs.

Aspect	Description	Technology approach
Strategic Context	Restructure organizational operations to adapt to modern challenges	Digital transformation strategy
Administrative Document Management	Digitize documents to optimize management processes	OCR (Tesseract), cloud platforms
Data Extraction & Processing	Convert scanned or printed documents into editable, searchable formats	OCR technology (Tesseract)
Overall Goal	Enable efficient information management and decision support	Integration of OCR, cloud computing, and digitization platforms

Table 1: Table of Digital transformation

In summary, the need for digital transformation in administrative and educational document management stems not only from the need to modernize and optimize processes, but also from the practical need for the ability to store, retrieve and process information quickly and accurately. The application of technologies such as OCR, cloud computing and data digitization platforms will be the key to realizing this goal.

2.2. Standard structure of Vietnamese administrative documents

According to Decree 30/2020/ND-CP of the Government of Vietnam [7], administrative documents are clearly regulated in terms of layout, presentation form and issuance process to ensure consistency in state management. The structure of the document includes main components such as the national emblem, motto, name of the issuing agency, document number and symbol, place name and date of issuance, summary of content, main content, signature and seal. In addition, the decree also specifically regulates the document format (font, font size, margin, line spacing), arrangement of components and rules for presenting appendices or attached documents. The input data of the system is Vietnamese administrative documents, for example as illustrated below:

BỘ GIÁO DỤC VÀ ĐÀO TẠO

Số: 06 /2016/TT-BGDĐT

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

Hà Nội, ngày 21 tháng 3 năm 2016

THÔNG TƯ
Ban hành Quy chế tuyển sinh đi học nước ngoài

Căn cứ Luật Giáo dục ngày 14 tháng 6 năm 2005; Luật sửa đổi, bổ sung một số điều của Luật Giáo dục ngày 25 tháng 11 năm 2009;

Căn cứ Luật Giáo dục đại học ngày 18 tháng 6 năm 2012;

Căn cứ Nghị định số 36/2012/NĐ-CP ngày 18 tháng 4 năm 2012 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của Bộ, cơ quan ngang Bộ;

Căn cứ Nghị định số 32/2008/NĐ-CP ngày 19 tháng 3 năm 2008 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của Bộ Giáo dục và Đào tạo;

Căn cứ Nghị định số 75/2006/NĐ-CP ngày 02 tháng 8 năm 2006 của Chính phủ quy định chi tiết và hướng dẫn thi hành một số điều của Luật Giáo dục; Nghị định số 31/2011/NĐ-CP ngày 11 tháng 5 năm 2011 của Chính phủ về việc sửa đổi, bổ sung một số điều của Nghị định số 75/2006/NĐ-CP ngày 02 tháng 8 năm 2006 của Chính phủ quy định chi tiết và hướng dẫn thi hành một số điều của Luật Giáo dục; Nghị định số 07/2013/NĐ-CP ngày 09/01/2013 của Chính phủ sửa đổi điểm b khoản 13 Điều 1 của Nghị định số 31/2011/NĐ-CP ngày 11 tháng 5 năm 2011 của Chính phủ về việc sửa đổi, bổ sung một số điều của Nghị định số 75/2006/NĐ-CP ngày 02 tháng 8 năm 2006 của Chính phủ quy định chi tiết và hướng dẫn thi hành một số điều của Luật Giáo dục;

Căn cứ Nghị định số 115/2010/NĐ-CP ngày 24 tháng 12 năm 2010 của Chính phủ quy định trách nhiệm quản lý nhà nước về giáo dục;

Theo đề nghị của Cục trưởng Cục Đào tạo với nước ngoài;

Bộ trưởng Bộ Giáo dục và Đào tạo ban hành Thông tư ban hành Quy chế tuyển sinh đi học nước ngoài.

Điều 1. Ban hành kèm theo Thông tư này Quy chế tuyển sinh đi học nước ngoài.

Điều 2. Thông tư này có hiệu lực kể từ ngày 06 tháng 5 năm 2016. Thông tư này thay thế Thông tư số 01/2013/TT-BGDĐT ngày 29 tháng 01 năm 2013 của Bộ trưởng Bộ Giáo dục và Đào tạo ban hành Quy chế tuyển sinh đi học nước ngoài.

Điều 3. Chánh Văn phòng, Cục trưởng Cục Đào tạo với nước ngoài, Thủ trưởng các đơn vị liên quan thuộc Bộ Giáo dục và Đào tạo; các tổ chức, cá nhân có liên quan chịu trách nhiệm thi hành Thông tư này./.

Nơi nhận:
- Văn phòng Quốc hội;
- Văn phòng Chủ tịch nước;
- Văn phòng Chính phủ;
- Ban Tuyên giáo TƯ;
- UBVHGDTNTNNĐ của QH;
- Các Bộ, cơ quan ngang Bộ, cơ quan trực thuộc CP;
- Cục KTrVBQPPL (Bộ Tư pháp);
- Kiểm toán nhà nước;
- Bộ trưởng (để b/c);
- Như Điều 3 (để thực hiện);
- Công báo;
- Website Bộ GDĐT;
- Lưu: VT, Vụ PC, Cục ĐTVNN.

KT. BỘ TRƯỞNG
THỨ TRƯỞNG

Bùi Văn Ga

Figure 2: Example of an administrative document.

(This form is a standard document type, including all important components and complying with the current regulatory format of the Vietnamese Government.)

The structure of a Vietnamese administrative document is shown in the following figure:

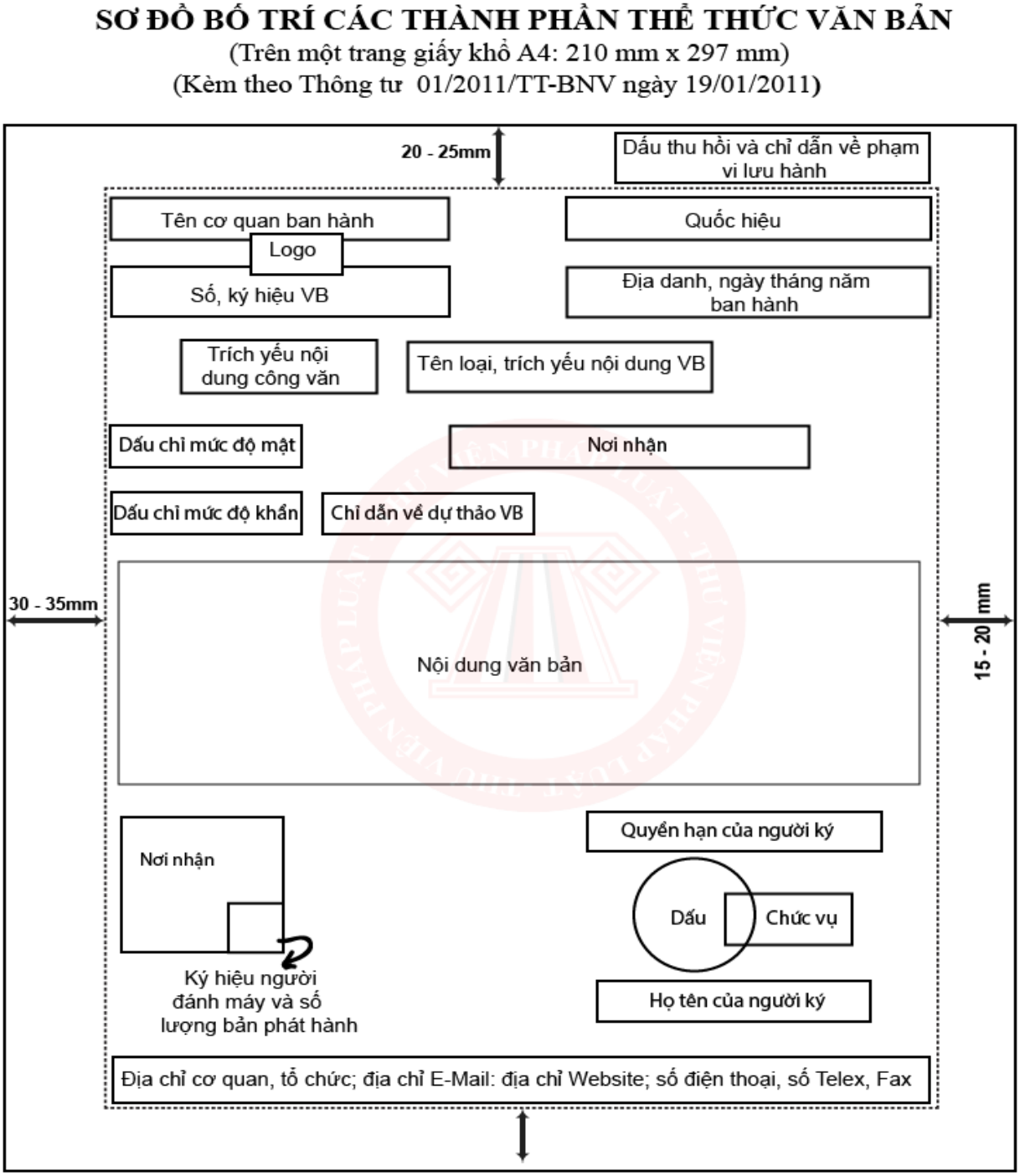


Figure 3: Structure format of an administrative document.

The scope of the research focuses on building, testing and evaluating the web application in a simulated and real environment, with data being common types of administrative documents in Vietnam. The system not only meets the organization's requirements in document management but also aims to expand the ability to integrate with other management systems in the future, serving the goal of comprehensive digital transformation.

2.3. Page segmentation and image preprocessing for OCR

2.3.1. Page Segmentation

Page segmentation is a fundamental step in document image analysis that aims to divide a page into meaningful and functionally distinct regions. Typical regions include text blocks, tables, figures, and annotations. Accurate segmentation is crucial for subsequent processes such as Optical Character Recognition (OCR) and information extraction, as it prevents misclassification of content and ensures higher processing accuracy [8].

Method	Technique	Step
Projection-Based Methods	Use pixel distribution along horizontal and vertical axes	1. Compute horizontal projection, detect text/image blocks by high pixel density 2. Compute vertical projection, identify columns or separated content 3. Use whitespace and lines to separate blocks
Smearing-Based Methods	Connect adjacent pixels to form larger blocks	1. Identify neighbouring pixels 2. Merge connected pixels to form blocks 3. Group characters into paragraphs or content areas
Connected Component-Based Methods	Build higher-level structures from groups of adjacent pixels	1. Identify connected pixels, each element is a character, symbol, or small object 2. Analyze spatial relationships between elements 3. Reconstruct larger content areas like lines, paragraphs, tables

Table 2: Table of Page segmentation method

According to Kise [8], page segmentation methods can be classified based on the analysis strategy and the primitives used. The three main approaches are Projection-Based Methods, Smearing-Based Methods, and Connected Component-Based Methods.

Projection-Based Methods

Projection-Based Methods are essential techniques utilized in page segmentation, falling under the broader category of Layout Analysis. These methods analyze the distribution of pixels along both horizontal and vertical axes, considering white space and lines to identify distinct content blocks on a document page. For example, in administrative documents, elements such as titles, main content, and signatures are typically spatially separated. Horizontal projection

helps pinpoint regions with high pixel density, which correspond to text or image blocks, while vertical projection aids in recognizing column layouts, particularly in tables.

Smearing-Based Methods

Smearing-Based Methods focus on connecting adjacent pixels to create larger content blocks, effectively merging individual characters or symbols into cohesive areas. This technique is particularly beneficial in dense text, as it integrates letters into a unified block representing entire paragraphs while distinguishing them from other elements like tables or images.

Connected Component-Based Methods

Connected Component-Based Methods involve identifying groups of adjacent pixels that serve as the foundation for constructing higher-level structures such as text lines, paragraphs, or tables. Each linked element can be a character, symbol, or small object. By examining the spatial relationships among these elements, it becomes possible to reconstruct larger content areas.

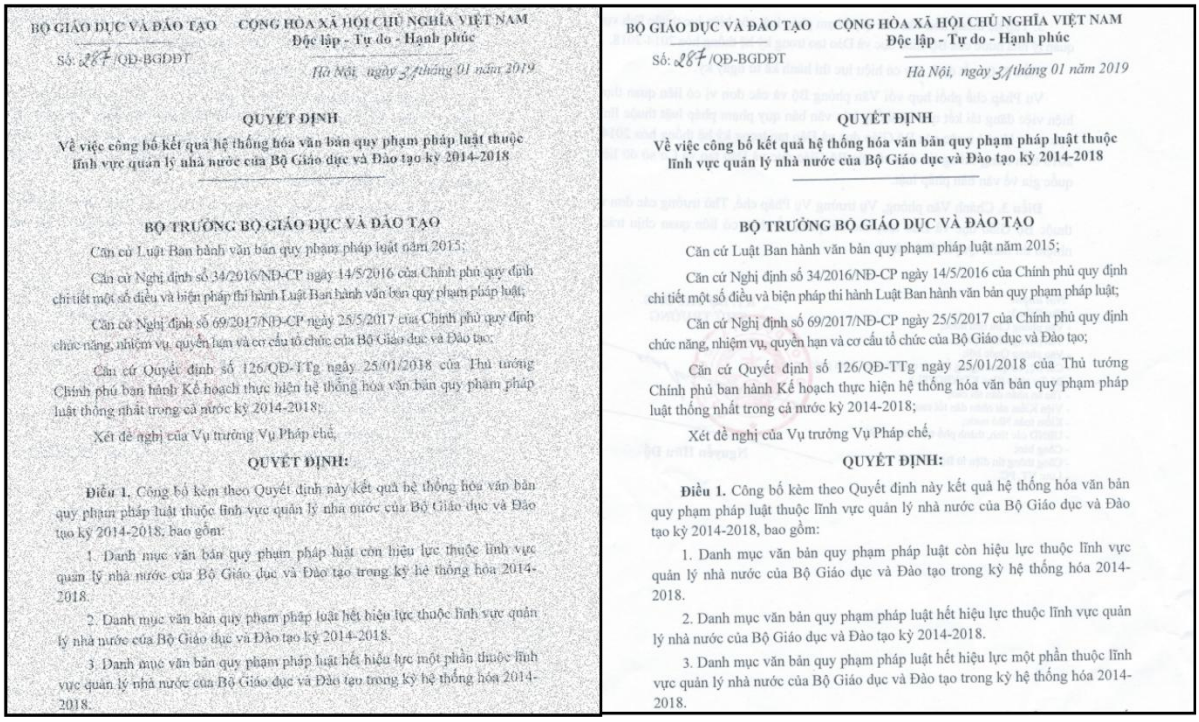
2.3.2. Image Preprocessing

After completing the page segmentation step, image preprocessing plays a key role in ensuring the quality of image regions before entering the optical character recognition (OCR) system. The goal of this step is to homogenize the data, remove noise factors, and increase the ability to distinguish characters, thereby improving the accuracy of OCR.

One of the most basic and common steps is converting the image to grayscale. Color documents usually have three RGB channels, while image processing and OCR algorithms are mainly based on light intensity. Converting to grayscale reduces the data from three color channels to a single channel, reducing the computational load and improving the stability of the algorithm, while preserving the brightness information of the pixels. In this way, the characters and lines in the document still retain the contrast, but the processing becomes faster and more efficient.

Noise removal is an im portance step in image preprocessing. Scanned documents, especially old paper or multiple photocopies, often contain small noise points that cause OCR to misrecognize characters. Filters such as Gaussian Blur or Bilateral Filter are applied to

remove noise while still maintaining sharp and clear text. Gaussian Blur helps smooth out random noise areas, while Bilateral Filter also preserves edge lines, preventing characters from being blurred or losing detail. As a result, letters become clearer and easier to recognize.



BỘ GIÁO DỤC VÀ ĐÀO TẠO
CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

Số: 287 /QĐ-BGDĐT

Hà Nội, ngày 31 tháng 01 năm 2019

QUYẾT ĐỊNH

Về việc công bố kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018

BỘ TRƯỞNG BỘ GIÁO DỤC VÀ ĐÀO TẠO

Căn cứ Luật Ban hành văn bản quy phạm pháp luật năm 2015;

Căn cứ Nghị định số 34/2016/NĐ-CP ngày 14/5/2016 của Chính phủ quy định chi tiết một số điều và biện pháp thi hành Luật Ban hành văn bản quy phạm pháp luật;

Căn cứ Nghị định số 69/2017/NĐ-CP ngày 25/5/2017 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của Bộ Giáo dục và Đào tạo;

Căn cứ Quyết định số 126/QĐ-TTg ngày 25/01/2018 của Thủ tướng Chính phủ ban hành Kế hoạch thực hiện hệ thống hóa văn bản quy phạm pháp luật thống nhất trong cả nước kỳ 2014-2018;

Xét đề nghị của Vụ trưởng Vụ Pháp chế,

QUYẾT ĐỊNH:

Điều 1. Công bố kèm theo Quyết định này kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018, bao gồm:

1. Danh mục văn bản quy phạm pháp luật còn hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

2. Danh mục văn bản quy phạm pháp luật hết hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018.

3. Danh mục văn bản quy phạm pháp luật hết hiệu lực một phần thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

Figure 4: Legal Document after Noise removal.

To improve the ability to distinguish letters in dark or dim areas of the image, light and dark equalization and contrast enhancement techniques are used, typically Histogram Equalization or CLAHE (Contrast Limited Adaptive Histogram Equalization). These methods adjust the intensity distribution of pixels in the image, highlighting letters against the background and reducing the influence of dark areas or uneven lighting, thereby supporting OCR to recognize more accurately.

Another common problem is the skewing of text (deskewing). During scanning or photographing, the document page is often tilted, causing the lines of text to be misaligned, leading to errors in OCR. The tilt angle can be determined by Hough Transform or analysis of the bounding box of the characters, and the image is then rotated to align the lines of text. This ensures that characters are correctly recognized line by line, minimizing errors caused by tilted or misaligned text.

In some special cases, such as when scanning bound books (book warp) or photographing documents with perspective distortion, it is necessary to perform Geometric Distortion

Correction. This step helps to remove bending or perspective distortions, ensuring that characters and content blocks retain their correct shape. This allows OCR and subsequent processing steps to operate accurately on the normalized input data.

In summary, image preprocessing includes grayscale conversion, noise reduction, light and dark balance and contrast enhancement, tilt correction and distortion correction, to ensure that the input image for OCR is consistent, clear and minimizes recognition errors. This is especially important for languages such as Vietnamese, which has many accents and compound characters, which are easily affected by low image quality or distortion during scanning [8], [9].

2.3.3. Cross-Domain Applications of Image Processing

Studies in areas such as 3D cultural heritage digitization and medical image processing have shown the key role of data normalization and distortion removal before proceeding to in-depth analysis steps.

In 3D cultural heritage digitization, Tran et al. [10] applied geometric noise filtering and 3D model calibration techniques to ensure that the original shapes and details are preserved, while removing noise or distortion. This helps the modeling and visualization process achieve high fidelity, better supporting conservation and research.

In medical image processing, Nguyen et al. [11] implemented pixel intensity normalization and Gaussian noise reduction to ensure that data from different devices or shooting conditions remain consistent. As a result, image analysis algorithms, such as tissue segmentation or pathology detection, achieve higher accuracy and minimize errors due to data source differences.

These experiences can be applied to legal document digitization, where input data often comes from a variety of sources such as paper scans, photographs, or original PDF files. Inconsistent quality and format can easily lead to errors in OCR and information extraction. Therefore, image standardization and distortion removal at the outset are key to ensuring accurate, consistent output that is ready for subsequent automated processing steps.

2.4. Natural language processing for legal documents

2.4.1. Training data requirements for legal NLP

To build an NLP model for legal documents, it is necessary to prepare a large enough, diverse dataset and label it appropriately for the task. Commonly used data types include:

- Full-text legal documents: constitutions, laws, decrees, circulars, decisions.
- Case precedents and judgments: trial documents, legal arguments of the court.
- Sample contracts and terms: serve the task of extracting terms, comparing documents.
- Administrative documents: official dispatches, notices, instructions on law application.

The training dataset for clause classification can include vary articles belonging to groups such as criminal, civil, commercial, labor. Each data sample will include the content of the article and the corresponding topic label.Data quality directly affects model results: data must be cleaned, language standardized, OCR errors eliminated, and legal structure (article numbers, clauses, points) preserved.

2.4.2. Popular methods and models implement for training data

Topic Modeling is a technique for discovering latent topic groups in a corpus based on word distribution. One of the classic models is Latent Dirichlet Allocation (LDA) proposed by Blei et al. [12]. LDA assumes that each document is a combination of many topics, and each topic is a probability distribution over words. This method is especially useful when it is necessary to discover groups of terms or legal documents with similar content, for example, a group of articles on "labor contracts" or "intellectual property rights".

However, LDA is based on the bag-of-words model, which does not take advantage of deeper semantic information from modern embedding. Dieng et al. [13] proposed Topic Modeling in Embedding Spaces (ETM), in which words and topics are represented in the same embedding space. This approach allows for the discovery of more semantically coherent topics, for example, recognizing that the words “divorce”, “marriage”, and “division of property” are closely related in the context of family law, even if they do not co-occur frequently in the same document.

For short text passages, such as legal title or judgment summary, traditional topic models often struggle due to the lack of contextual information. BERTopic [14] overcomes this problem by combining the embedding representation from the Transformer model (e.g., BERT) with the UMAP dimensionality reduction technique and HDBSCAN clustering. The system then applies Class-based TF-IDF to extract keywords representing each topic cluster.

While the original BERT was trained on general language, Chalkidis et al. [15] developed LEGAL-BERT, a variant pretrained on a large legal corpus, including laws, contracts, and case law. LEGAL-BERT is optimized to capture legal language, including specific sentence structures, specialized terminology, and legal expression patterns.

PhoBERT [16] is a pre-trained language model based on the BERT architecture, optimized for Vietnamese, using a corpus of more than 20GB of text. PhoBERT has good semantic representation capabilities, supporting tasks such as text classification, entity tagging (NER), and dependency parsing.

VnCoreNLP [17] is an NLP toolkit for Vietnamese, including word segmentation, part-of-speech (POS) tagging, entity recognition, and parsing. This toolkit helps standardize input data for language models, which is especially important when processing legal documents that contain many compound words, proper names, and legal terms.

A. Tewari [18] introduces LegalPro-BERT, a BERT model specifically fine-tuned for the task of classifying legal clauses. Different from general text classification, this task requires the model to deeply understand the meaning of the clause, including references to articles, clauses, and other legal documents.

2.5. OCR technology and development

2.5.1. Overview of OCR

Optical Character Recognition (OCR) is a technology that converts images containing text into character data that computers can search, edit, and process automatically. This technology plays an important role in the process of digitizing and exploiting text data, with a wide range of applications such as:

- Digitizing archived documents: Converting printed or handwritten documents into digital data for preservation and search.
- Automatic information extraction: From invoices, receipts, administrative forms.

- Supporting machine translation: Recognizing text from images and converting to another language.
- Legal document processing: Converting laws, decrees, contracts, and court records into structured data, allowing for quick searching and analysis.

While OCR for printed text has achieved very high accuracy, OCR for handwritten text (Handwritten OCR) [19] is still a big challenge due to the diversity of fonts, spacing, strokes and digital image quality.

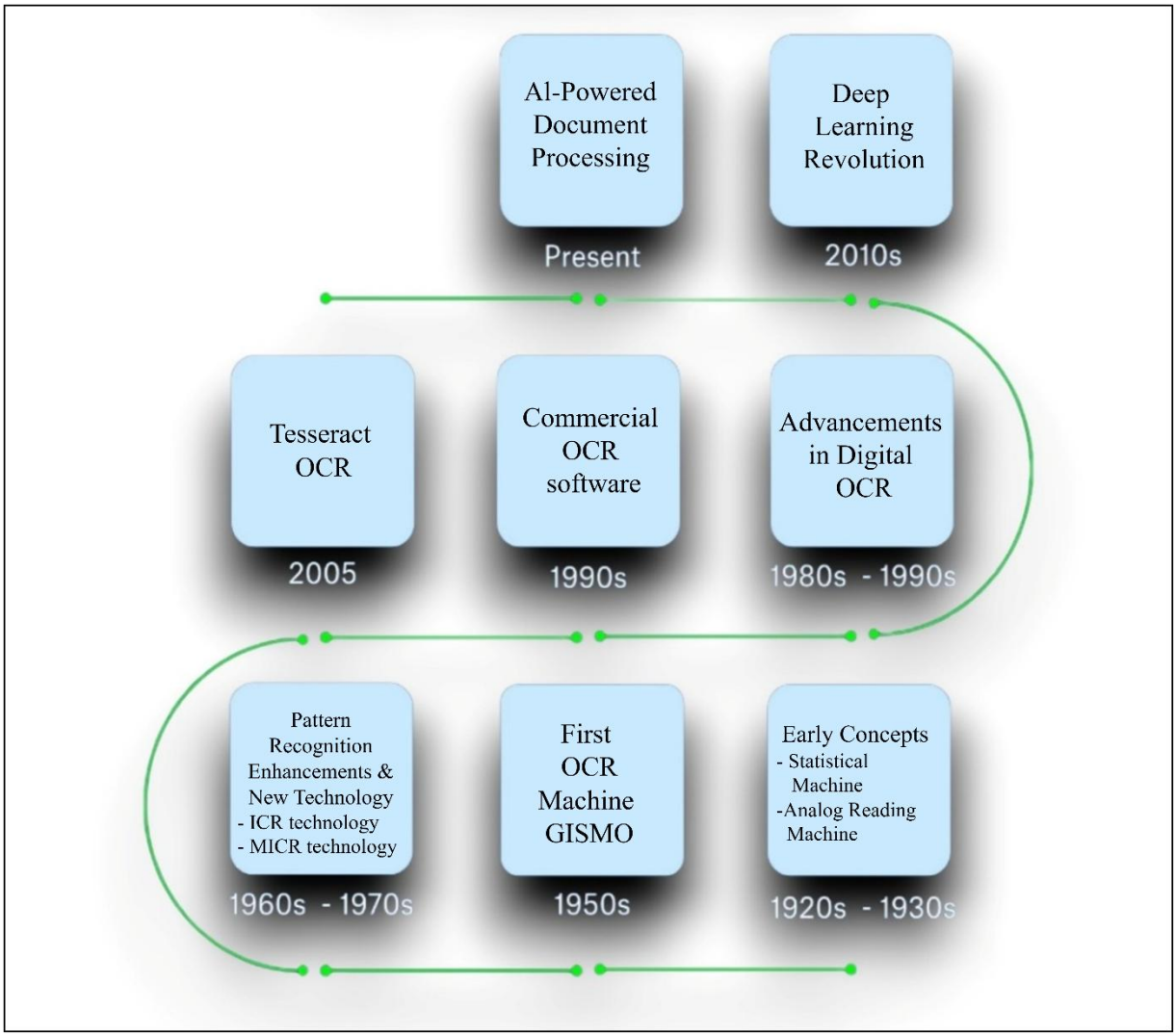


Figure 5: The development of OCR.

Due to the explosion of deep learning, especially convolutional neural networks (CNN) and recurrent networks (RNN, LSTM), OCR has achieved very high accuracy, especially with printed text. Modern models such as CRNN or TrOCR combine computer vision and natural language processing, allowing for multilingual recognition and complex layouts.

2.5.2. Practical assessment using OCR

Although OCR technology has achieved very high accuracy in many applications, especially with clear print, when deployed in real environments, there are still many complex challenges. These limitations come not only from the characteristics of the input data but also from the processing algorithms and conditions of use:

Unrecognizable handwriting

Handwritten OCR (H-OCR) is one of the most difficult problems of OCR [19]. The reason lies in the large differences in shape, tilt, size, and thickness of the strokes between individuals. Handwriting often has cursive writing, uneven spacing, and is sometimes shortened or distorted compared to the standard model. Errors in distinguishing characters that are easily confused (for example, “1” and “l”, “n” and “r”).

Poor input image quality

OCR requires input images with high enough resolution and low noise to extract accurate features [21]. However, in practice, documents are often affected by factors such as blurring due to hand shake, noise due to degraded old paper, distortion due to shooting at an angle, uneven lighting or shadows.

Difficult segmentation

Segmentation is the step of separating a document into logical units such as text regions, lines, words, and characters. For documents with simple layouts, this step can be performed using traditional techniques such as projection histograms or connected components [24]. However, documents with many columns, tables, illustrations, or text placed on the background image will make segmentation more complicated.

Complex layout

Many legal documents, technical documents or administrative forms have a non-linear structure: titles, notes, tables, columns, and item numbers are presented interwoven. In addition, special symbols (e.g. mathematical symbols, special characters) are often not in the standard OCR character set.

2.5.3. Tesseract OCR for Vietnamese

Tesseract is one of the most popular open source OCR tools, developed by Hewlett-Packard in the mid-1980s and later maintained and improved by Google [20]. This tool supports many languages, including Vietnamese. For Vietnamese, the biggest challenge lies in the complex system of accents and accented characters. Tesseract has overcome this problem by using a machine learning model based on recurrent neural networks (LSTM) in new versions, allowing more accurate recognition of accented characters and improved reliability compared to traditional OCR methods.

In practice, Tesseract can achieve good performance when processing printed documents with clear fonts, but the accuracy decreases when encountering blurred, noisy text or complex layouts. Additional training (fine-tuning) on a specific Vietnamese dataset can significantly improve performance.

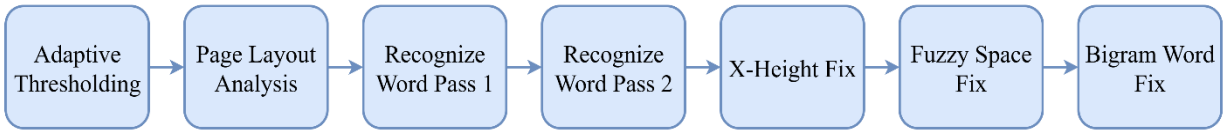


Figure 6: Block diagram of the overall architecture of Tesseract [20]

The overall architecture of Tesseract (built on a pipeline model) has been maintained stable for many years. The system consists of the following main components:

Adaptive Thresholding

- The first step is to binarize the image (convert from grayscale to black-and-white).
- Helps separate text from the background by changing the threshold for each region, instead of applying a fixed threshold.

Page Layout Analysis

- Identify text blocks, lines, words, and characters in the document.
- This is an important step for OCR to understand the structure of the original document.

Recognize Word - Pass 1 & Pass 2

- Pass 1: Use the character classifier to recognize and simultaneously train the context-adaptive classifier while processing.

- Pass 2: Return to recognize difficult or uncertain words in Pass 1, based on the adapted classifier.
- This two-pass approach significantly improves accuracy.

Fixing Stages

- X-Height Fix: Fixes errors related to character height.
- Fuzzy Space Fix: Fixes word spacing (when OCR is uncertain about white space).
- Word Bigram Fix: Uses linguistic context (pairs of consecutive words) to fix recognition errors.

2.5.4. Hand-writing text OCR for Vietnamese

Handwriting OCR (handwriting recognition) is still a much more difficult problem. The reason comes from the diversity of writing styles, slant, stroke thickness, size and individual differences between writers. These variations make traditional techniques based on hand-crafted features less effective, as they are not capable of generalizing to handle the rich diversity in handwritten data [33]. The development of Deep Learning, especially convolutional neural networks (CNNs) and recurrent neural networks (RNNs), has opened up a new direction for Handwriting OCR. Instead of manual character segmentation and feature extraction, deep learning models can directly learn features from input data and process character sequences automatically. Among them, Convolutional Recurrent Neural Network (CRNN) is considered a prominent architecture, thanks to its ability to combine the power of CNN in image feature extraction and RNN in character sequence modeling, along with the Connectionist Temporal Classification (CTC) loss function that eliminates the need for precise character label alignment [34]. CRNN was introduced by Shi et al. (2016) in the context of scene text recognition and was quickly applied to Handwriting OCR [34][35]. This architecture consists of three main components:

Convolutional Neural Network (CNN) - Image Feature Extraction

CNN is used as an automatic feature extractor from handwritten images. Convolutional layers and pooling layers help the model learn spatial features such as lines, edges, and shapes of each character. Thanks to the local invariance property, CNN is especially effective in processing handwritten text that has many variations in position, tilt, or stroke thickness [36].

The output of CNN is not the original image, but a series of deep features, considered as "feature maps" that describe each horizontal slice of the text. This series of features is the input to RNN.

Recurrent Neural Network (RNN) - Character Sequence Modeling

After the CNN extracts the features, the RNN (usually using Bi-directional LSTM) takes this feature sequence to model the sequential relationship. This is important because handwriting is not a set of independent characters but a continuous sequence, where the context of the previous character affects the recognition of the next character. BiLSTM is especially useful because it considers both the forward and backward context, which helps improve the accuracy of text string recognition [37].

Connectionist Temporal Classification (CTC) Loss - Labeling without segmentation

In traditional models, to train OCR, it is necessary to align each character in the label with the correct position in the image. However, handwriting is highly variable, making it difficult to manually label each position. CTC loss (Graves et al., 2006) solves this problem by allowing the model to generate a longer candidate character sequence than the actual label by inserting "blank tokens". CTC then automatically calculates the probability of all possible alignments and optimizes according to the final label sequence [38]. This allows CRNN to learn directly from data without segmentation, a breakthrough in Handwriting OCR.

Overall mechanism

The working process of CRNN can be described as follows:

- Input handwritten image is fed into CNN → generates feature sequence.
- Feature sequence goes through RNN (BiLSTM) → models sequential relations, forming a representation vector for each time step.
- Softmax layer assigns probability to each character at each time step.
- CTC loss optimizes the whole process by comparing the predicted sequence with the final label sequence.

The strength of the CRNN architecture is its end-to-end recognition capability, meaning that the entire process from input image to output text string is learned in a unified model, without the need for manual character segmentation (segmentation-free). This makes CRNN particularly effective when dealing with free-form handwriting.

2.6. Techniques for OCR recognition

2.6.1. Thresholding for robust binarization

Thresholding is an important preprocessing step in the OCR pipeline, especially for Vietnamese, where the input documents often include old printed papers, multiple photocopies, or photos taken from phones with uneven lighting conditions. A classic review by Sezgin and Sankur [22] systematized thresholding methods into three main groups: global thresholding, adaptive/local thresholding, and statistical optimization methods such as Otsu.

Global Thresholding: Determine a single threshold value for the entire image. Otsu's method [22] is a typical representative, based on the principle of minimizing the intra-class variance or maximizing the inter-class variance of the image intensity histogram. This method is particularly effective when the data has a clear bimodal distribution between the background and the text.

Adaptive Thresholding: Suitable for images with non-uniform lighting or contrast, by calculating a threshold for each image region based on the average or median value in the neighboring window. Algorithms such as Sauvola or Niblack improve the ability to separate text in areas with shadows, ink smears, or uneven backgrounds [22].

Vietnamese OCR context: In practice, when processing legal documents, contracts, or administrative records, Vietnamese scanned images often encounter many problems such as yellowing paper, fading ink, patterned backgrounds, or red seals overlapping the text. Selecting and fine-tuning the binarization method (e.g., combining Otsu for the homogeneous background and adaptive thresholding for the shadow region) can significantly reduce the downstream character recognition error, especially for complex accented characters such as "ă", "â", "đ", "ễ" which are sensitive to edge noise.

In addition, some recent studies have proposed multi-level Otsu variants and combined them with morphological filtering to improve the sharpness of characters before feeding them into Tesseract or VietOCR, helping to improve the recognition accuracy in the Vietnamese character set.

2.6.2. Detecting circles using Hough Circle Transform

In many types of administrative documents, contracts or certificates in Vietnam, circular seals play an important role as a legal authentication factor. The near-perfect geometrical characteristics of the seals allow the application of Hough Circle Transform for accurate detection in the parameter space, which is an extension of the traditional Hough Transform for circles. Hough Circle Transform maps each edge point in the image to a three-dimensional space consisting of center (x, y) and radius r, then finds the maximum in this space to identify potential circles [23]. Parameters such as dp (cumulative spatial image resolution ratio), minDist (minimum distance between detected centers), param1 and param2 (thresholds for edge and maximum detection), along with minRadius and maxRadius (circle size limit) play a key role in balancing detection sensitivity and noise reduction [23].

Step	OpenCV Parameter	Process
1. Input Preprocessed Image	Grayscale / Edge Image	The image is converted to grayscale and edges are detected (e.g., using Canny) to highlight circular boundaries and reduce noise, preparing it for Hough Circle Transform.
2. Apply Hough Circle Transform	cv2.HoughCircles()	Maps each edge point to a 3D parameter space (center x, y and radius r) and finds maxima to identify potential circles, leveraging the near-perfect geometry of seals.
3. Set Transform Parameters	dp, minDist, param1, param2, minRadius, maxRadius	Controls detection sensitivity and noise reduction: dp for resolution ratio, minDist for minimum center distance, param1/param2 for edge and detection thresholds, minRadius/maxRadius to constrain circle size. Fine-tuning improves detection accuracy for diverse document scans.
4. Output Detected Circles	List of circles (x_center, y_center, radius)	Provides coordinates and size of each detected seal, enabling either preprocessing to remove seal regions before OCR or extracting seals for authentication, classification, or legal storage.

Table 3: Table of Hough Circle process

In the context of Vietnamese OCR, detecting and processing seals can serve two main purposes:

- Preprocessing to remove the seal area before character recognition, to avoid motifs, characters or patterns in the seal from interfering with the OCR algorithm.
- Extracting the seal as a separate entity for authentication, document classification or legal sign storage.
- Research and implementation practice show that, with Vietnamese administrative document data that is diverse in scan quality, color and contrast, fine-tuning the Hough Circle Transform parameter for each specific data set helps to significantly improve detection accuracy, especially when the seal is blurred, tilted or overprinted on the text content.

2.6.3. Find contours to separate text layout blocks

In document image processing, contours are an important tool to identify and separate information-containing regions, especially useful in documents with many columns, tables or annotation blocks. According to the OpenCV manual [24], the contour detection process usually includes the following main steps:

Image preprocessing - The input image is usually converted to grayscale, applying binarization techniques (thresholding) or edge detection (edge detection) such as Canny or Sobel to highlight the boundaries between regions.

Step	OpenCV Function / Parameter	Process
1. Input Preprocessed Image	Grayscale / Binarized / Edge Image	The image is converted to grayscale and optionally binarized or edge-detected (Canny, Sobel) to highlight boundaries and reduce noise, preparing it for contour detection.
2. Call cv2.findContours()	contours, hierarchy = cv2.findContours(image, mode, method)	Detects all continuous boundaries of objects or text blocks in the image, providing the basic geometric shapes for further processing.
3. Set Retrieval Mode	mode parameter: - RETR_EXTERNAL - RETR_TREE	Determines which contours are extracted and how hierarchical relationships are stored: RETR_EXTERNAL for outermost blocks (e.g., columns), RETR_TREE for all contours with parent-child info (e.g., table cells, annotations).
4. Set Approximation Method	method parameter: - CHAIN_APPROX_SIMPLE - CHAIN_APPROX_NONE	Defines how contour points are stored: CHAIN_APPROX_SIMPLE compresses points to reduce memory, while CHAIN_APPROX_NONE keeps all points for full detail.
5. Output	contours, hierarchy	Provides the list of detected contours and hierarchical info, representing text/layout blocks ready for OCR or further document analysis.

Table 4: Table of Contours to separate layout blocks

Find contours with OpenCV - The cv2.findContours() function allows to detect all contours based on the processed image region. This process is controlled by two important parameters:

Retrieval Mode:

- RETR_EXTERNAL: only takes the outermost contour, ignoring nested contours, useful when only major blocks such as text columns need to be extracted.
- RETR_TREE: takes the entire contour and stores the hierarchy, allowing for parent-child relationships between blocks, useful when cells in tables or comments are extracted within large blocks.

Approximation Method:

- CHAIN_APPROX_SIMPLE: compresses the number of stored contour points (keeps corner points, discards collinear points), reducing memory footprint and speeding up processing.

Applications in Document OCR - With scanned or imaged text data, contours support:

Precise text block separation: Locate columns, paragraphs, or table areas before feeding each area into OCR, avoiding misrecognition due to interleaved content.

Exploit hierarchical information: From RETR_TREE mode, it is possible to identify child contours that are contained within a parent contour, such as table cells, checkboxes, or annotation areas.

Reduce confusion with non-text elements: Remove patterns, decorative borders, or watermarks, focusing OCR on the part containing characters.

2.6.4. Classification and recognition using Support Vector Machines (SVM)

Support Vector Machines (SVM) is one of the most widely used and sustainable supervised learning algorithms in the field of classification and recognition. In the context of LegalTech, SVM has demonstrated its ability to effectively combine high accuracy, good generalization ability, and reasonable computational cost compared to many more complex models.

In principle, SVM searches for an optimal hyperplane to separate data belonging to different classes, maximizing the distance (margin) between this hyperplane and the nearest data points (support vectors). This feature helps SVM perform well even when the data has high-dimensional data, which is a characteristic of legal text processing problems.

Applications in legal document classification

In modern legal decision support systems, SVM is often used for:

- Classifying legal documents (laws, decrees, contracts, precedents).
- Predicting case outcomes based on historical data.
- Optimizing information retrieval in large legal databases.

According to BytePlus [25], SVM can be deployed in server infrastructure and provided via API, allowing direct integration into online legal analysis systems. A study in the Perm Krai region (Russia) [26] developed a legal document classification system based on SVM, deployed it in a real server environment, and demonstrated its effectiveness in both processing speed and accuracy.

In paper [26], the legal document classification system built on SVM followed the regional classification used in Perm Krai. The list of classes defined in this work includes 20 main legal subject groups. This list is the basis for the SVM system to classify legal documents, ensuring fast, accurate search and compliance with regional requirements. Classes with uneven data distribution, such as “Labor and Employment” or “International Affairs”, therefore the author had to consider solutions such as adding data from other regions or using imbalance-robust models to maintain the accuracy of the system.

Notably, in a comparative study on the LexGLUE benchmark dataset [27], SVM achieved competitive performance compared to transformer models such as BERT. Although BERT is generally superior in accuracy, the difference in the legal context is not as large, especially when training and deployment costs are considered. This suggests that SVM is still the optimal choice in many cases where stable performance and rapid deployment are required.

Applications in character recognition

In addition to language processing, SVM is also widely applied in character recognition, especially handwritten characters. For example, on Bangla handwritten digit datasets such as NumtaDB, CMARTdb, Ekush, and BDRW, the combination of Histogram of Oriented Gradients (HOG) and SVM features achieved an accuracy [28]. This success comes from the ability of SVM to separate complex data classes based on character shape and structure, making it a reliable solution when dealing with handwritten or signed legal documents, where high accuracy and the ability to handle diverse character variations are required.

CHAPTER 3: METHODOLOGY

3.1. Overview

3.1.1. Proposed method

The system operates in a sequential process designed to ensure the accuracy and integrity of the output data. First, PDF documents are converted into high-quality images that retain the details and sharpness of each page, which serve as the basis for character recognition. Next, these images undergo preprocessing, including noise removal, brightness adjustment, and contrast enhancement, to improve the efficiency of OCR technology in extracting accurate content. After preprocessing, the system detects and classifies text areas on the page, distinguishing between printed and handwritten letters using Support Vector Machines (SVM), and then selects the appropriate character recognition method for each type. For printed letters, the system uses Tesseract OCR for recognition, while handwritten letters are processed using Convolutional Recurrent Neural Network (CRNN), which helps to recognize characters with diverse fonts and difficult to read. After character recognition, the text is normalized and reconstructed according to the original document layout, including content ordering, adjusting whitespace, punctuation, and special characters. Finally, the processed data is converted to a structured XML format, and text elements such as headings, paragraphs, tables, or images are labeled, creating an output that is easy to manage and can be directly integrated into automated data storage or analysis systems.

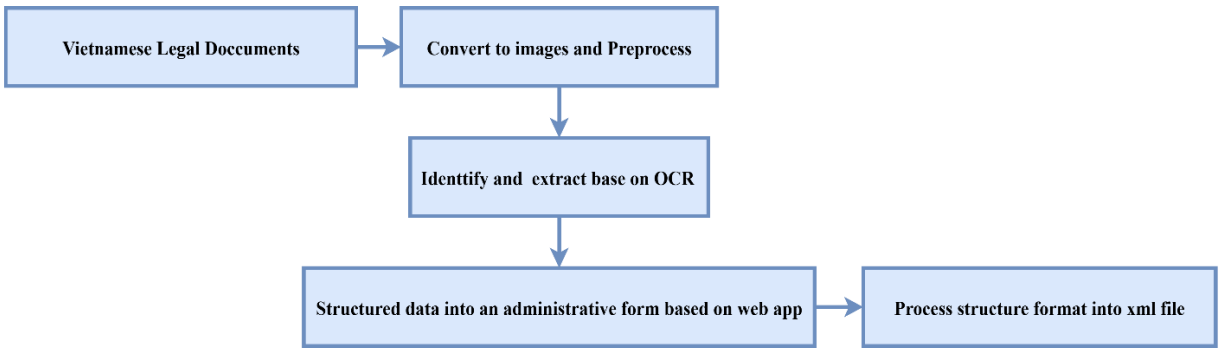


Figure 7: Proposed workflow for legal text processing and tagging

3.1.2. Platform Overview

The document recognition and processing system is built on the Python platform, combining powerful open source libraries in image processing, machine learning and OCR such as OpenCV (image preprocessing), scikit-learn (machine learning algorithm), TensorFlow/PyTorch (training and deploying Vietnamese handwriting recognition model) and Tesseract OCR (printed text recognition). The system architecture is designed in separate modules, ensuring that each component can be maintained, upgraded or replaced independently without interrupting the entire processing process.

Specifically, the Flask application acts as a web interface layer, receiving input files from users (PDF, images) and coordinating the processing flow through the modules:

- **pdf_converter:** Convert PDF to high-quality input image sequence.
- **ocr_engine:** Call the OCR model to extract text from images. The model uses the architecture from the open source project Vietnamese-Handwriting-Recognition-OCR [GitHub] with the ability to recognize Vietnamese handwriting, including variations of accents and features.
- **text_saver**: Save recognition results to multiple output formats (XML, JSON, CSV, HTML) to meet the needs of data storage, exchange or integration into automatic document management/data analysis systems.
- **loading_waiting**: Process signals and provide status feedback during the time the system performs recognition.

This design makes the system easy to expand (for example, add OCR models for other languages or improve the image preprocessing pipeline) and adapt to diverse document processing needs in real environments. By combining deep learning models and traditional algorithms, the system can process both printed and handwritten letters with high accuracy, especially suitable for administrative, legal and historical documents.

3.2. Collect data

The data source is collected from the official website of the Ministry of Education and Training of Vietnam (MOET). This is the official electronic information portal, providing a repository of legal documents and administrative documents related to the education sector, including decrees, circulars, decisions and official dispatches. These documents are often stored in scanned PDF files, containing both text content and illustrative images.

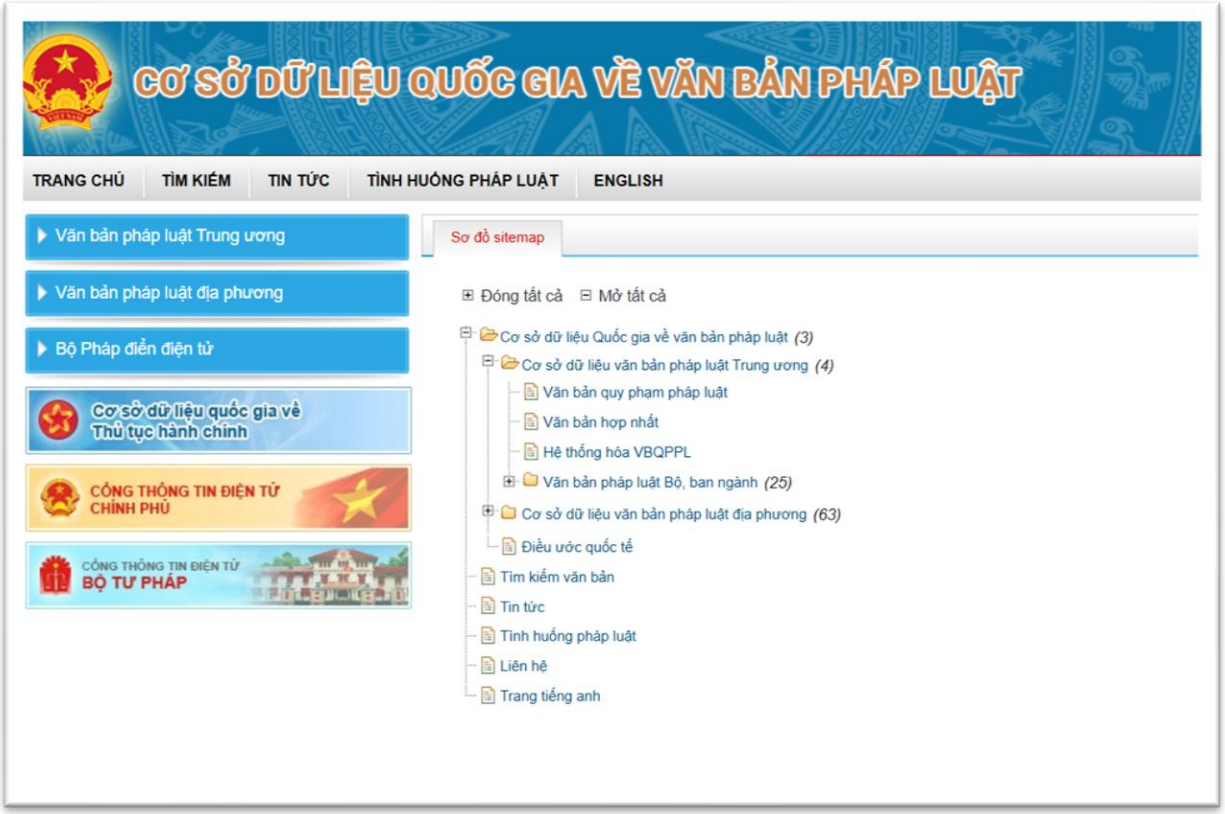


Figure 8: National Database of Legal Documents Sitemap

To serve the research, the team built an automatic data collection algorithm written in Python, which allows the system to periodically access the website, search for new documents and download them to the server. This approach not only ensures the timeliness and completeness of the data but also minimizes the risk of errors compared to the manual collection process.

After downloading, each PDF file will be converted page by page to a raster image (PNG or JPG format) with high resolution, to ensure the input quality for the next image processing steps. The image processing process includes techniques such as contrast enhancement, noise

reduction and grayscale conversion to highlight the text area. This is an important step to significantly improve the efficiency of optical character recognition (OCR) in the next stage.

After recognizing the text, the system proceeds to structure the data into administrative information fields, and finally converts it to XML format for storage, searching and integration into legal document management systems.

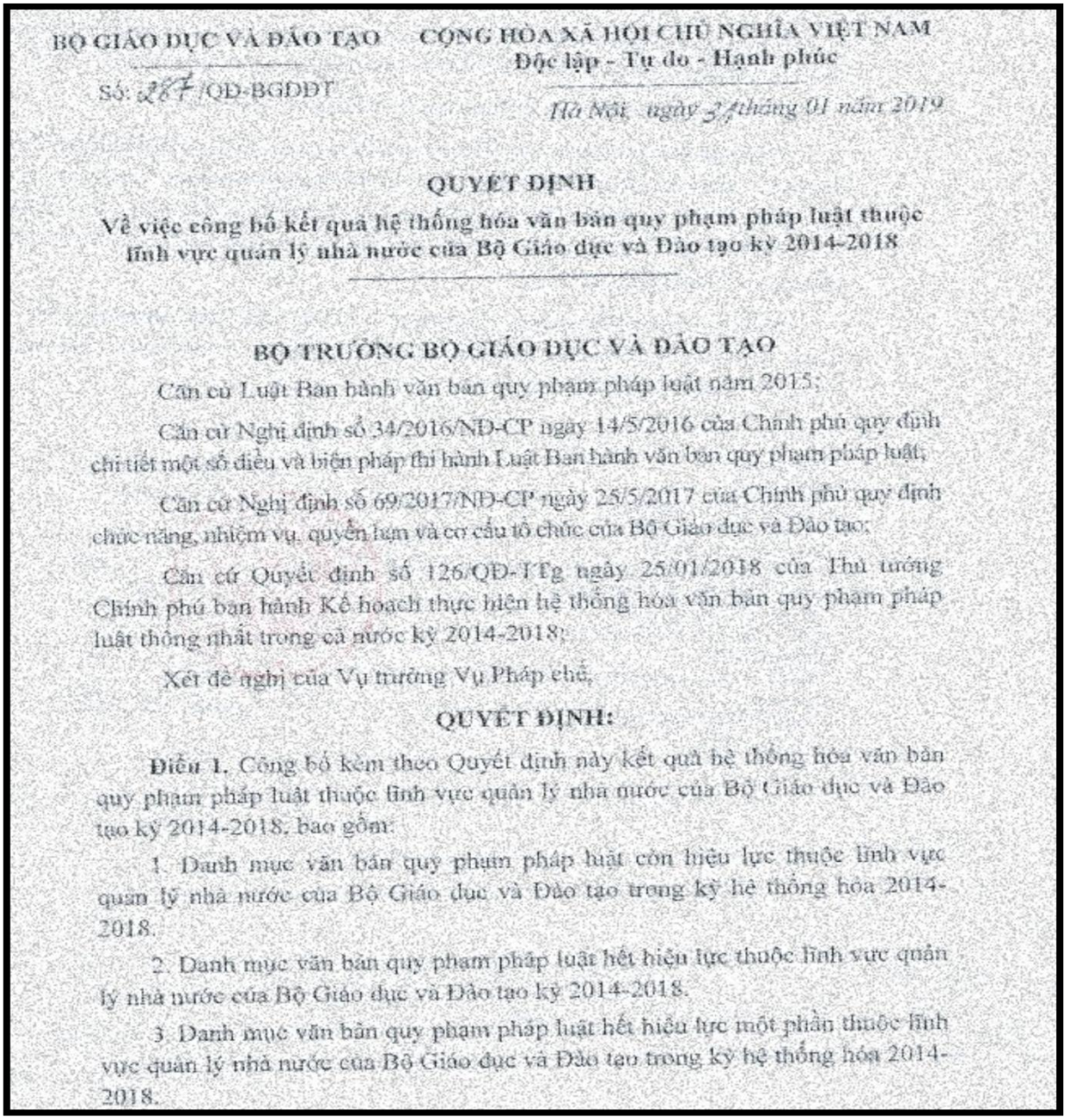
Below is algorithm for detailed description of the research process in this thesis introducing the above content.:

Algorithm of DataCollection():

```
Input: base_webpage_url of the MOET website
Output: Set of PDF files downloaded
Set current_page_number = 1
Set base_site_url = "https://moet.gov.vn/van-ban/?Page="
while current_page_number is not NULL do:
    Visit: base_site_url + current_page_number
        // Example: https://moet.gov.vn/van-ban/?Page=1
    Parse the HTML content of the current page
    For each element in the HTML content:
        If the element text contains "Bản PDF":
            Extract the `href` link of the PDF file from the element
                // Example href: "/TW/Pages/vbpq-van-ban-goc.aspx?ItemID=..."
            Download the PDF file
        End if
    End for
    If "next page" link is found in the page:
        current_page_number++
    End if
End while (when current_page_number = NULL )
Return: "All files downloaded successfully"
```

3.3. Image processing

One of the importance factors that determines the accuracy of the OCR system is the quality of the input image. Due to the characteristics of legal documents published on the official portal of the Ministry of Education and Training, many documents exist in scanned PDF format with uneven quality: some pages have high blur, a lot of background noise or are degraded due to the digitization process. Therefore, the initial image preprocessing step plays a key role in standardizing the data and improving the quality of character display before entering the recognition module.



BỘ GIÁO DỤC VÀ ĐÀO TẠO

Số: 287 /QĐ-BGDĐT

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

Hà Nội, ngày 31 tháng 01 năm 2019

QUYẾT ĐỊNH

Về việc công bố kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018

BỘ TRƯỞNG BỘ GIÁO DỤC VÀ ĐÀO TẠO

Căn cứ Luật Ban hành văn bản quy phạm pháp luật năm 2015;

Căn cứ Nghị định số 34/2016/NĐ-CP ngày 14/5/2016 của Chính phủ quy định chi tiết một số điều và biện pháp thi hành Luật Ban hành văn bản quy phạm pháp luật;

Căn cứ Nghị định số 69/2017/NĐ-CP ngày 25/5/2017 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của Bộ Giáo dục và Đào tạo;

Căn cứ Quyết định số 126/QĐ-TTg ngày 25/01/2018 của Thủ tướng Chính phủ ban hành Kế hoạch thực hiện hệ thống hóa văn bản quy phạm pháp luật thống nhất trong cả nước kỳ 2014-2018;

Xét đề nghị của Vụ trưởng Vụ Pháp chế,

QUYẾT ĐỊNH:

Điều 1. Công bố kèm theo Quyết định này kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018, bao gồm:

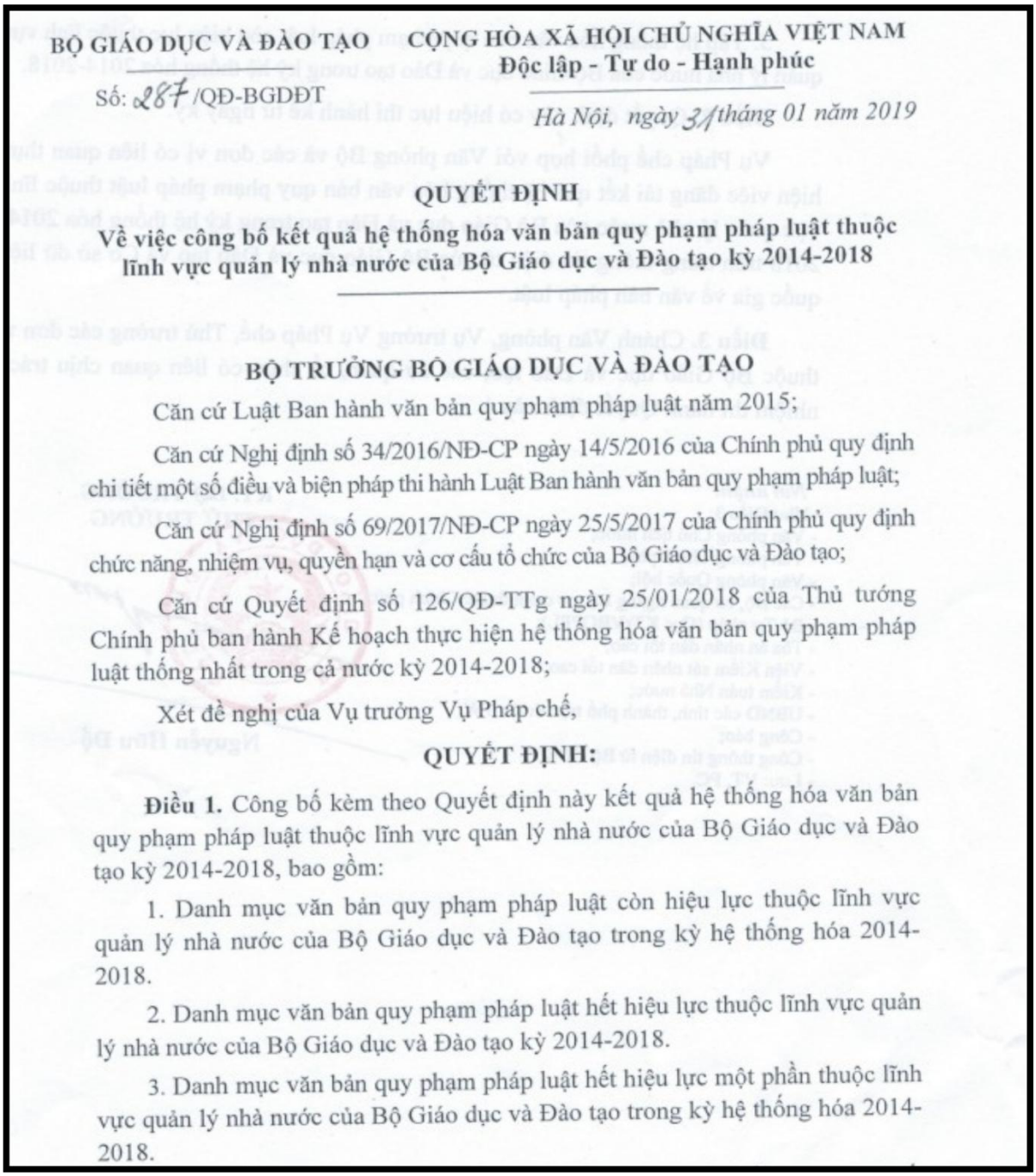
1. Danh mục văn bản quy phạm pháp luật còn hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

2. Danh mục văn bản quy phạm pháp luật hết hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018.

3. Danh mục văn bản quy phạm pháp luật hết hiệu lực một phần thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

Figure 9: The original image of legal documents included many noise

In this thesis, the two most important techniques applied are denoising and thresholding. Denoising is considered a necessary first step to remove redundant details that affect the recognition process. Scanned images often appear in many different types of noise, from random noise to salt and pepper noise. Image filtering algorithms such as Gaussian filter and Median filter are applied flexibly: while Gaussian filter helps smooth the whole image and reduce small noise particles, Median filter is especially useful when preserving character contours, preventing the strokes from being "softened" too much. Thanks to that, the character shape becomes clearer, keeping the original structure but eliminating most of the noise.



BỘ GIÁO DỤC VÀ ĐÀO TẠO

Số: 287 /QĐ-BGDĐT

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

Hà Nội, ngày 31 tháng 01 năm 2019

QUYẾT ĐỊNH

Về việc công bố kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018

BỘ TRƯỞNG BỘ GIÁO DỤC VÀ ĐÀO TẠO

Căn cứ Luật Ban hành văn bản quy phạm pháp luật năm 2015;

Căn cứ Nghị định số 34/2016/NĐ-CP ngày 14/5/2016 của Chính phủ quy định chi tiết một số điều và biện pháp thi hành Luật Ban hành văn bản quy phạm pháp luật;

Căn cứ Nghị định số 69/2017/NĐ-CP ngày 25/5/2017 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của Bộ Giáo dục và Đào tạo;

Căn cứ Quyết định số 126/QĐ-TTg ngày 25/01/2018 của Thủ tướng Chính phủ ban hành Kế hoạch thực hiện hệ thống hóa văn bản quy phạm pháp luật thống nhất trong cả nước kỳ 2014-2018;

Xét đề nghị của Vụ trưởng Vụ Pháp chế,

QUYẾT ĐỊNH:

Điều 1. Công bố kèm theo Quyết định này kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018, bao gồm:

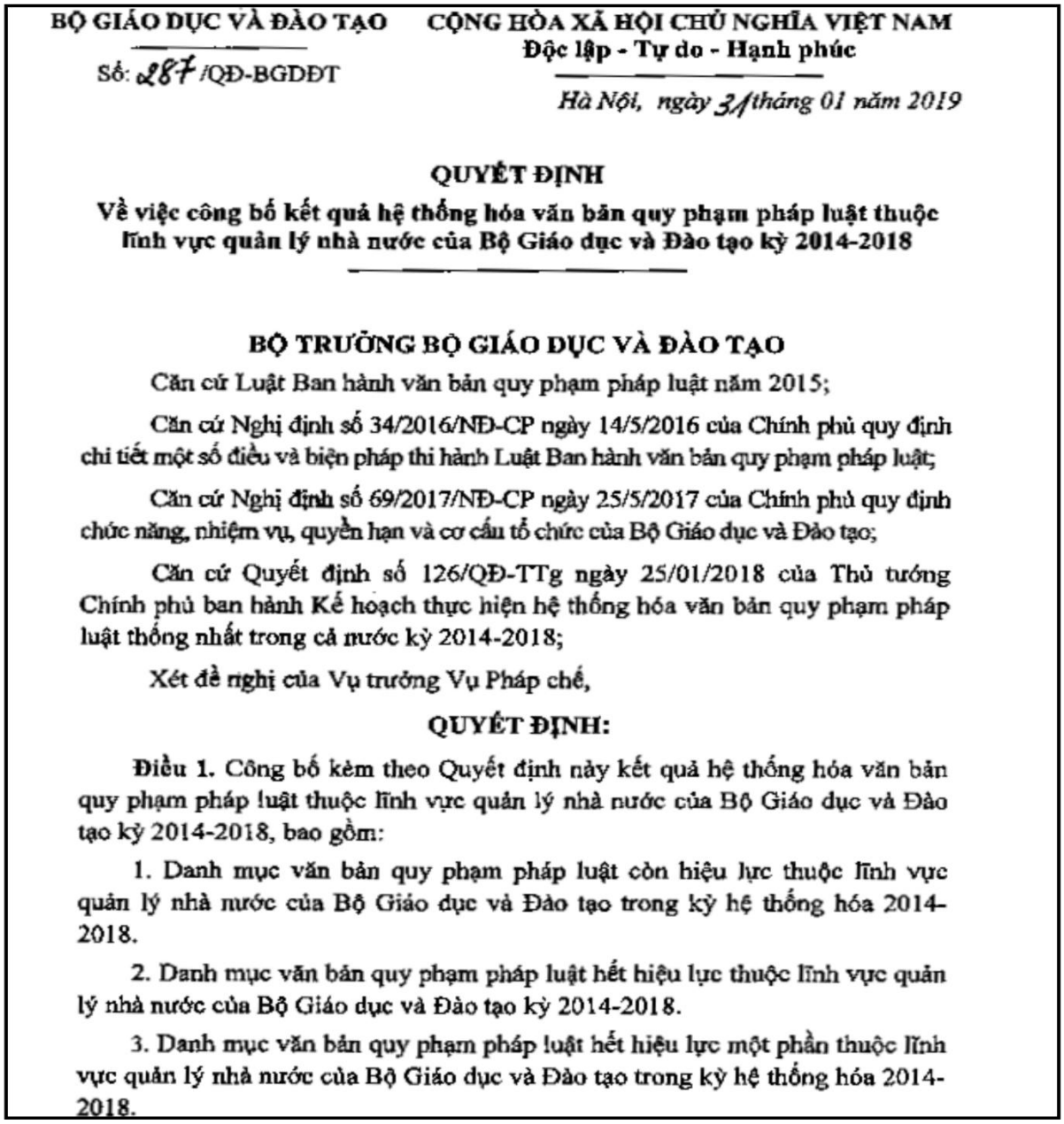
1. Danh mục văn bản quy phạm pháp luật còn hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

2. Danh mục văn bản quy phạm pháp luật hết hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018.

3. Danh mục văn bản quy phạm pháp luật hết hiệu lực một phần thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

Figure 10: The image after Denoising

After the denoising step, the image continues to be put into the binarization stage (thresholding). This is a technique to convert the image to black-and-white, in which the pixels are classified into background and text areas. Binarization is important because most OCR models work better on binary data than on grayscale or color images. In this study, adaptive thresholding methods are applied instead of just using a fixed threshold. This is particularly useful in the context of legal documents with inconsistent scan quality: the same page can contain both overexposed and underexposed areas. Adaptive thresholding allows for small areas of the image to be analyzed to determine the appropriate threshold, which allows for more accurate separation of text from the background.



BỘ GIÁO DỤC VÀ ĐÀO TẠO

Số: 287 /QĐ-BGDĐT

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

Hà Nội, ngày 31 tháng 01 năm 2019

QUYẾT ĐỊNH

Về việc công bố kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018

BỘ TRƯỞNG BỘ GIÁO DỤC VÀ ĐÀO TẠO

Căn cứ Luật Ban hành văn bản quy phạm pháp luật năm 2015;

Căn cứ Nghị định số 34/2016/NĐ-CP ngày 14/5/2016 của Chính phủ quy định chi tiết một số điều và biện pháp thi hành Luật Ban hành văn bản quy phạm pháp luật;

Căn cứ Nghị định số 69/2017/NĐ-CP ngày 25/5/2017 của Chính phủ quy định chức năng, nhiệm vụ, quyền hạn và cơ cấu tổ chức của Bộ Giáo dục và Đào tạo;

Căn cứ Quyết định số 126/QĐ-TTg ngày 25/01/2018 của Thủ tướng Chính phủ ban hành Kế hoạch thực hiện hệ thống hóa văn bản quy phạm pháp luật thống nhất trong cả nước kỳ 2014-2018;

Xét đề nghị của Vụ trưởng Vụ Pháp chế,

QUYẾT ĐỊNH:

Điều 1. Công bố kèm theo Quyết định này kết quả hệ thống hóa văn bản quy phạm pháp luật thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018, bao gồm:

1. Danh mục văn bản quy phạm pháp luật còn hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

2. Danh mục văn bản quy phạm pháp luật hết hiệu lực thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo kỳ 2014-2018.

3. Danh mục văn bản quy phạm pháp luật hết hiệu lực một phần thuộc lĩnh vực quản lý nhà nước của Bộ Giáo dục và Đào tạo trong kỳ hệ thống hóa 2014-2018.

Figure 11: Thresholding the image to separate the text in image

These two processing steps complement each other: denoising helps clean up the image, while binarisation simplifies and highlights the recognized object. Experiments have shown that combining both techniques significantly increases the OCR accuracy, especially for documents of medium or poor quality. Additional steps such as converting to grayscale or adjusting contrast can still be performed, but denoising and thresholding are the key factors that improve the efficiency of legal document processing.

3.4. Text Recognition and Extraction

3.4.1. Advanced Preprocessing for OCR

Advanced preprocessing plays an important role in improving the quality of input images, thereby improving the accuracy of optical character recognition (OCR). In addition to basic techniques such as noise filtering or skew correction, some advanced methods are applied to optimize the ability to separate text from the background and maintain the character structure.

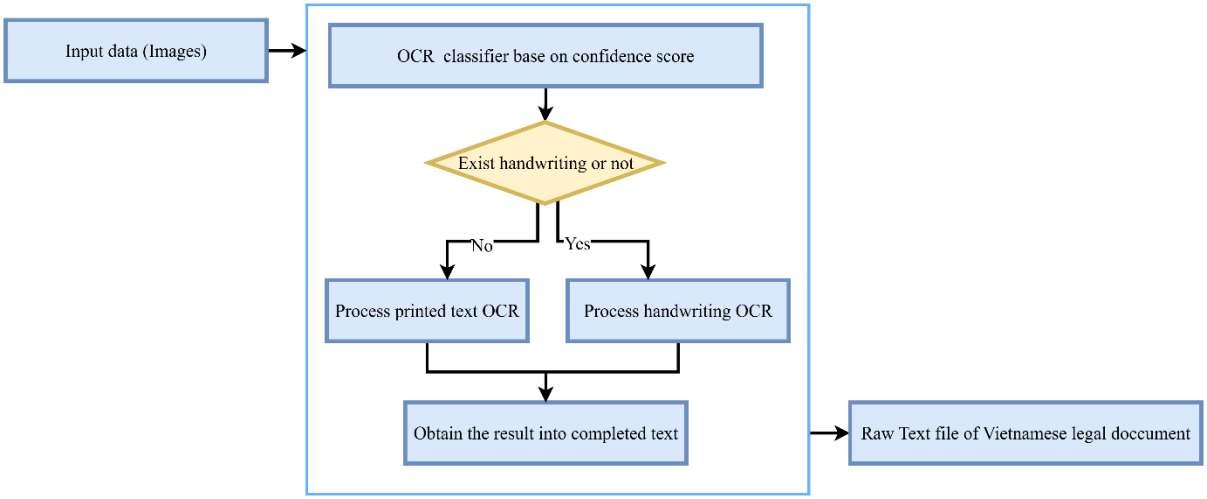


Figure 12: OCR processing for extracting raw text from pdf document file.

To effectively extract text from scanned legal documents, the system employs a hybrid OCR pipeline that begins with region segmentation and classification. After the document is pre-processed and divided into individual text regions, each segment undergoes OCR processing using Tesseract. In addition to recognizing the text content, Tesseract outputs a confidence score for each word, which reflects the reliability of its prediction.

These confidence scores are then used to classify the text as either printed or handwritten. High-confidence text is typically associated with printed material, whereas handwritten text tends to yield lower scores due to irregularities in shape and structure. This classification mechanism

allows the system to dynamically select the most appropriate OCR approach for each word region.

- Printed text is processed using Tesseract OCR (v4.1.1), which includes support for Vietnamese and performs well on structured, machine-printed fonts.
- Handwritten text, such as annotations or signatures, is passed to a Convolutional Recurrent Neural Network (CRNN) model

3.4.2. Text Region Classification Using SVM

In the system architecture, the clear recognition of regions containing printed and handwritten text plays a key role. This allows the system of appropriate recognition models for each type: Tesseract OCR for printed text, and deep learning models such as CRNN for handwritten text. To perform this classification, the system uses Support Vector Machines (SVMs) with geometric features extracted from the text region, to accurately distinguish between the two types of characters.

Method implementation

- The geometric and structural features of the text region are selected based on the ability to clearly distinguish between printed and handwritten text:
- Curvature of the strokes: Handwritten text often shows soft curves or irregular corners, unlike sharp edges in printed text.
- Pixel density: Printed text often has a high and relatively uniform pixel density, while handwritten text has an uneven distribution of points, in some places the strokes can be thin or thick.
- Character alignment and layout: Printed text is usually arranged in uniform rows and columns, while handwritten text can be skewed, tilted, or not follow a fixed format.

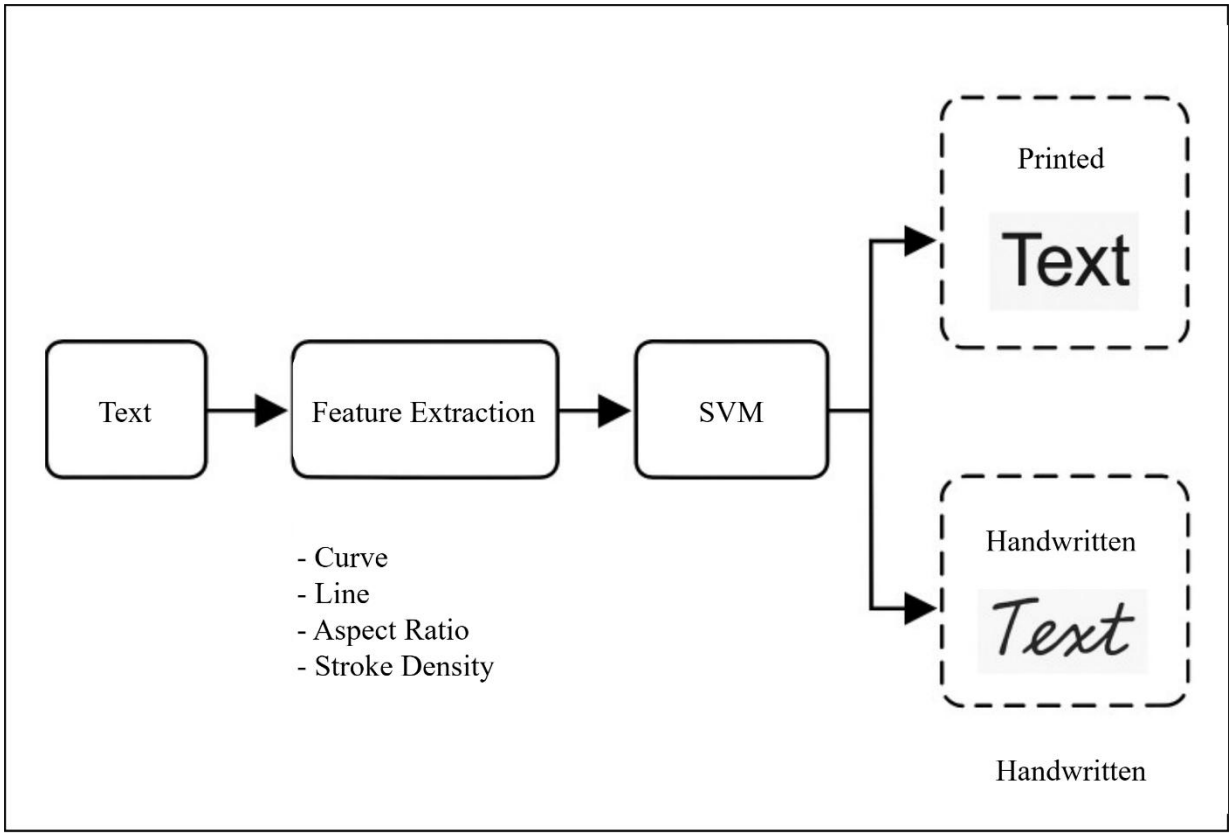


Figure 13: SVM to separate the handwriting-text and printed-text

After collecting these features from each text segment, the system uses SVM to train a binary classifier based on the label: printed text and handwritten text. The goal is to build an optimal hyperplane that can distinguish the two classes with a wide enough margin to increase generalization ability, while reducing errors when encountering new typefaces.

3.4.3. Print-Text Recognition Using Tesseract OCR

In the text recognition and digitization system, Tesseract OCR (Optical Character Recognition) is chosen as the main tool to process and extract printed text content from Vietnamese legal documents. This is an open source library, originally developed by Hewlett-Packard and later maintained by Google, famous for its multilingual recognition capabilities and flexibility in configuration.

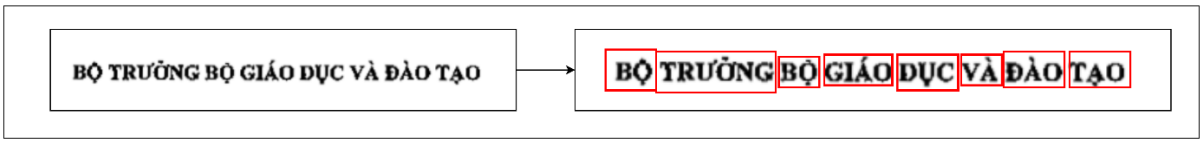


Figure 14 Print Recognition Output Using Tesseract OCR

Document layout analysis (Layout Analysis):

- Separate text from the background and non-text components (images, lines, tables).
- Identify blocks of text, lines and words.

Character Recognition:

- Use recurrent neural networks (LSTM - Long Short-Term Memory) to recognize character strings.
- Integrating a language model to predict and correct recognized text strings, reducing errors at the word and sentence levels.
- Configuration for Vietnamese legal documents

Applying Tesseract OCR to Vietnamese faces many challenges due to:

- A large number of tone marks (sharp, flat, question, fall, heavy).
- The quality of legal documents is often affected by the printing and scanning process.

To overcome this, the system is configured with:

- Vietnamese training data set (vie.traineddata) developed and optimized by the community.
- Print recognition mode (OEM and PSM): Use --oem 1 (LSTM OCR Engine) combined with --psm 6 (assuming the document has many lines).
- Customize the dictionary (word list and user patterns): Add specialized legal terminology to increase accuracy when recognizing documents with specific language.

3.4.4. Handwriting-Text Recognition Using CRNN

In handwriting recognition, one of the major challenges is the diversity of writing styles, sizes, slant, and connections between characters. To solve this problem, CRNN (Convolutional Recurrent Neural Network) is considered an effective architecture thanks to its ability to combine spatial feature extraction and temporal sequence relationship modeling.

CRNN is a combination of three main components:

- **Convolutional layer (CNN):** CNN is used to extract local features from handwritten images. Convolutional and pooling layers help reduce noise, retaining important information about lines, curvature, and character shape.
- **Recurrent layer (RNN - usually using LSTM or GRU):** After the spatial features are extracted, RNN takes on the role of modelling the sequential dependencies between horizontal features of the text image. Thanks to that, the system can recognize the continuity and context between characters in handwriting.
- **CTC (Connectionist Temporal Classification) layer:** Since the input data is an image with variable length and the character labels are not precisely aligned over each time step, CRNN uses the CTC loss function to map the feature string to the output character string. CTC allows the model to learn directly without aligning each character in the image.

Advantages of CRNN in handwriting recognition:

- Ability to handle data with variable length, suitable for natural handwriting.
- Automatically learn features from raw image data, independent of manual rules.
- Achieve higher accuracy than traditional methods when applied to complex handwriting datasets.

Component (Layer)	Output Shape	Parameters	Main Function	Illustrative Explanation
Input Layer	(118, 2167, 1)	0	Input is a preprocessed grayscale handwriting image.	Example: image of size 118×2167 pixels.
Conv2D (64 filters)	(118, 2167, 64) → pooling	640	First CNN layer that extracts low-level features such as edges.	Detects vertical and horizontal strokes of characters.
MaxPooling2D	(39, 722, 64)	0	Reduces the size of the feature map.	Keeps only essential features while discarding noise.
Conv2D (128 filters)	(39, 722, 128) → pooling	73,856	Extracts more complex features such as curves and loops.	Helps differentiate between letters like “c” and “o”.
Conv2D (256 filters) + BatchNorm	(13, 240, 256)	295,168	Learns mid-level representations of handwriting shapes.	Character contours become clearer.

Residual Block (Add + Activation)	(13, 240, 256)	-	Skip connections preserve original information while deepening the model.	Prevents information loss in deeper layers.
Conv2D (512 filters) + BatchNorm + Residual	(13, 240, 512)	1,180,160 + 2,048	Extracts higher-level, complex handwriting features.	Helps distinguish visually similar letters like “i” and “j”.
Conv2D (1024 filters)	(13, 240, 1024) → pooling	4,719,616	Provides strong representation across the entire text region.	Clearly separates each character in a word.
Lambda (reshape)	(240, 1024)	0	Reshapes feature maps into a sequence of feature vectors.	Each vector corresponds to a vertical slice of the image.
Bi-LSTM (2 layers, 512 units each direction)	(240, 1024)	6,295,552 × 2	Models sequential dependencies in both forward and backward directions.	Understands character context, e.g., “th” usually appears together.
Dense Layer (Softmax)	(240, 141)	144,525	Converts feature vectors into probability distributions over classes.	141 output classes include letters, digits, and symbols.
CTC Loss Layer	(1,)	0	Optimizes alignment between predicted sequences and ground truth text.	Solves mismatch between image length and text length.

Table 5: Table of Handwriting-Text Recognition CRNN layer

In the proposed system, CRNN is chosen for the handwriting recognition stage to take advantage of deep learning capabilities to model the diversity and complexity of handwriting data. This significantly improves recognition performance compared to methods that rely solely on CNNs or traditional techniques.

3.4.5. Structured Data into an Administrative Form

After completing the text recognition process, all extracted content is reassembled according to its original position in the source document. This reconstruction ensures that the final text maintains the structure and order of the original document, which is critical for subsequent processing tasks such as clause extraction, legal analysis, or data archiving.

The structuring step transforms raw text into machine-readable, semantically tagged data that can be visualized and edited through a web-based interface.

The structured content is then automatically populated into a dynamic web form, where each field corresponds to an XML tag generated in the previous step. This interface allows legal professionals to verify and adjust the extracted data, ensuring that it accurately reflects the original document's structure and intent.

Below is algorithm for detailed description of the research process in this thesis introducing the above content.:

Algorithm : OCRProcessor

```
1: Input: PDF legal documents
2: Output: Full legal text
3: Convert PDF files into high-resolution images
4: Initialize text_output
5: for each page_image do
6:       regions = DetectTextRegions(page_image) // using SVM
7:       for each region in regions do
8:           features = ExtractVisualFeatures(region) //using OCR
9:           if IsPrintedText(features) then
10:              text = PrintedText
11:          else
12:              text = HandwritingText
13:      end if
14:          Append text to text_output in proper order
15:   end for
16: end for
27: Return text_output
```

3.5. Convert into XML structure

3.5.1. Conversion objective

Label tagging is an important step in the process of digitizing legal documents, allowing the extraction of meaningful components from raw data (in .txt format) and reorganizing them into a structured format, specifically XML. The goal of this step is to identify, classify, and represent information units such as document numbers, articles,abstracts, and legal contexts to serve data mining, semantic search, and automated legal analysis tasks.

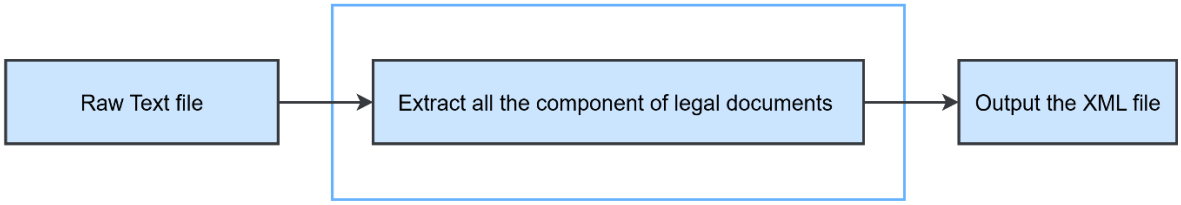


Figure 15: Processing for convert to XML

The plain legal text—obtained after OCR and normalization—is structured into a machine-readable XML format. This transformation is driven by a keyword-based mapping algorithm that identifies and segments critical components of Vietnamese legal documents.

The conversion process ensures that the document's original legal structure is preserved while enhancing its usability for further computational processing such as search, indexing, and data mining. Once the matching is complete, the tagged content is serialized into a well-formed XML document.

The main objective of converting to XML format is to clearly label each element in the document, helping the data to be systematically organized and easy to process. This standardization brings many core benefits, including important goals that any data management system must aim for:

- Long-term and structured storage: XML allows data to be stored in a standard, easy-to-preserve form, ensuring that no meaning is lost when retrieving later.
- Easy to visualize: Labeled data can be converted to many different representations (tables, charts, trees, visual interfaces) for analysis or presentation.

- Create a foundation for AI/ML model training: Structured and clearly labelled data is a mandatory requirement when used for machine learning systems, especially in natural language processing (NLP) problems.
- Data integration and exchange capabilities: XML is a common and standardized format, allowing data to be shared between different systems without worrying about losing compatibility.
- Enhanced search and retrieval capabilities: Thanks to labelling, users or systems can easily identify, search, and extract the right part of the desired information instead of having to process the entire raw text.
- Ensure accuracy and automation: Labelling and standardization help eliminate ambiguity, support efficient automated processing processes, and minimize human errors.

The process of converting to XML is not simply a format change, but also a fundamental step to ensure that data is tightly managed, effectively exploited, and well served for both storage, analysis, and artificial intelligence technology system.

3.5.2. Labeling Components

During the process of converting legal documents into XML format, each important component of the document is labelled with a corresponding tag. This labelling helps the data become structured, easy to retrieve, analyze, and integrate with other systems. Here are the main components:

Components	**Description**	**XML Tags**
1. quoc_hieu	National Title (e.g. Socialist Republic of Vietnam)	<quoc_hieu>
2. co_quan_ban_hanh	Issuing organization	<co_quan_ban_hanh>
3. ky_hieu_van_ban	Document ID	<ky_hieu_van_ban>
4. ngay_ban_hanh	Date of issuance	<ngay_ban_hanh>
5. loai_van_ban	Type of document	<loai_van_ban>

6. trich_yeu_noi_dung	Abstract or summary	<trich_yeu_noi_dung>
7. noi_dung_chinh	Main content	<noi_dung_chinh>
8. noi_nhan	Received organization (Recipient)	<noi_nhan>
9. nguoi_ky_van_ban	Full name and Signature of competent person	<nguoi_ky_van_ban>

Table 6: XML tags for structured legal document components

The structured content is automatically populated into a dynamic web form. Each field corresponds to one of the XML tags above and allows for visual review. This interface helps legal professionals ensure that the extracted data accurately mirrors the document's original structure and intent.

This approach facilitates better readability, efficient querying, and interoperability with legal databases and information systems, laying the foundation for advanced legal analytics and digital archiving.

3.5.3. Conversion Process

The final step in the process of digitizing legal documents is to convert raw data (plain text) into XML structure. Labelling text elements and mapping them to XML tags plays a particularly important role, as it allows:

- Extracting meaningful information from the original data.
- Reorganizing content according to a standardized structure.
- Enhancing the ability to store, search, exploit semantic data and serve automatic legal analysis system.

This conversion process is performed based on the keyword-based mapping algorithm. The input text is scanned, then the segments that match the pre-defined structural pattern (for example: "Number:…", "Article…", "Regarding…", or time expressions such as "day… month… year…") will be recognized and classified. After extraction, each element will be mapped to the corresponding XML tag.

In the process of digitizing legal documents, converting raw data (plain text) into structured XML is the final and important step.

- XML not only stores the content, but also adds “tags” to each element in the document.
- The system can understand the document not only as a string of characters, but as a set of data with legal meaning: number, clause, title, date of issue, issuing agency, etc.

This brings benefits:

- Extracting meaningful information (semantic information): instead of just reading the document, we can query like “get the entire Article 5 of document No. 06/2016/TT-BGDĐT”.
- Standardizing the structure: all legal documents follow a unified model (for example: <Article>, <Clause>, <Date>), making it easy to store and analyze.
- Automatic mining, searching and analysis: thanks to XML, legal analysis systems can automatically check the relevance between laws, compare effective dates, or detect conflicts between documents.

Below is algorithm for detailed description of the research process in this thesis introducing the above content.:

Algorithm of XMLconvert()

1: **Input**: Plain legal text
2: **Output**: Structured XML file
3: Extract all components of the legal document
4: **If** keywords match extracted segments
5: Map matched components to corresponding XML tags
6: **Convert** the structured content into a valid XML document

CHAPTER 4: IMPLEMENTATION AND RESULTS

4.1. Technology platform implement

The system is built on the Python programming language platform, because Python owns a rich ecosystem with many specialized libraries for image processing, optical character recognition (OCR) and natural language processing (NLP). In addition, modern web technologies are also integrated to provide a friendly interface and scalability. Specifically, the main technologies and libraries used include:

Image processing library:

To ensure the quality of the input data for the grammar evaluation system, the image preprocessing step plays an important role in the declarative development phase. The two main Python libraries used are OpenCV and Pillow to perform the necessary image processing tasks.

OpenCV is integrated to apply advanced image processing techniques, including noise removal to reduce impurities in the image, brightness and contrast adjustment to improve clarity, as well as text skew detection and correction to ensure that the text is edited accurately, serving the process of receiving more effective character recognition (OCR).

Pillow is used to perform additional basic image processing operations for OpenCV, limiting cropping and resizing images according to system requirements, and converting image formats to appropriate standards for subsequent processing steps.

The combination of OpenCV in the advanced tasks and Pillow in the operational base has created an efficient image data processing process and ensured stability for the entire system.

OCR Library:

Tesseract OCR was used as an open source optical character recognition tool. This tool is optimally configured for Vietnamese, helping to accurately recognize printed characters. Tesseract supports multiple text layout analysis modes, allowing for processing of documents with complex formats. In addition, the system also customizes Tesseract's dictionary to improve recognition accuracy, especially with specialized texts or documents containing specific vocabulary.

PDF extraction library:

PyMuPDF and pdf2image were used to convert legal documents in PDF format into high-quality raster images. This step ensures that the input data is qualified for subsequent processing, especially optical character recognition (OCR) and text analysis. Accurate conversion and high resolution retention help reduce recognition errors and optimize the efficiency of natural language processing (NLP) algorithms in the following steps.

Web Framework:

Flask is used to build the backend API, which plays a central role in coordinating the system's logic. This component is responsible for receiving requests from the user interface, connecting and coordinating activities between important processing modules such as OCR for character recognition, NLP for parsing and evaluating grammar, along with the module that converts data to XML format. After processing, Flask ensures that the results are returned to the frontend to display to the user. This architecture makes the system flexible, easy to integrate and expand in the future.

The system operates according to the client-server model:

Frontend (Flask) is responsible for providing the user interface, allowing operations such as loading documents, monitoring the processing process, and displaying results after analysis. The interface is designed to be friendly and intuitive to ensure that users can easily interact with the system without requiring in-depth technical knowledge.

Backend (Flask) takes on the role of processing the system's business logic. This component receives requests from the interface, coordinates the operations of important modules such as OCR for character recognition, NLP for language analysis, and performs data storage. After processing, the backend will return the results to the frontend to display to the user.

This architecture helps ensure scalability, easy maintenance and convenience when integrating new modules in the future, such as supporting handwritten text recognition or integrating a semantic search system.

4.2. System implement

The proposed system was implemented as a **web-based system** for digitizing Vietnamese legal and administrative documents. Its main functionalities are:

1. **Detecting** whether the content in an uploaded PDF document is printed text or handwriting.

2. **Extracting raw text data** using OCR (with Vietnamese language support).

3. **Structuring the recognized content** into XML format with correct tagging based on a predefined schema.

The system architecture consists of two primary modules:

1. **Document Processing Module** - PDF documents are first converted into high-resolution images using the pdf2image library in Python. The images undergo preprocessing techniques such as noise removal, binarization, and skew correction to improve OCR accuracy. Handwriting and printed text regions are classified separately, allowing the system to apply optimized OCR configurations for each type of text.

2. **Output Generation Module** - The recognized text is cleaned, segmented according to the legal document's structure, and mapped to predefined XML tags such as <quoc_hieu>, <co_quan_ban_hanh>, <ngay_ban_hanh>, and <noi_dung_chinh>. The final XML output complies with the required format for downstream system such as legal text search and NLP model training.

This implementation ensures an intuitive user experience and provides platform independence, as the tool is accessible from any modern web browser without requiring local installation.

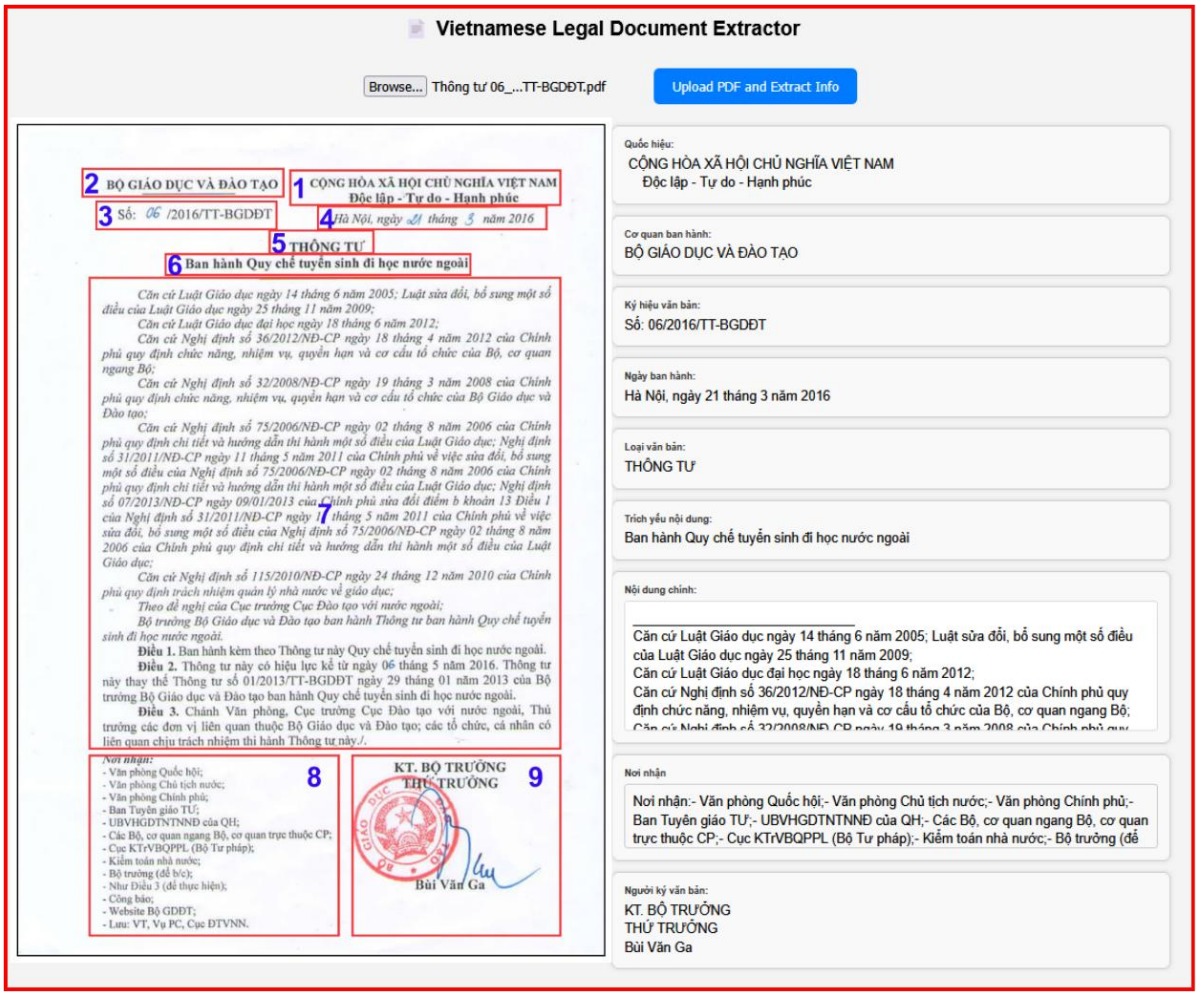


Figure 16: The web-app for implementing all steps of proposed method

The proposed legal document digitization system is deployed as a web-based system, designed to provide a simple and intuitive user experience. The interface is divided into three main sections:

1. **Document Upload Panel** - Located at the top of the page, this section allows users to upload Vietnamese legal or administrative documents in PDF format. Drag-and-drop functionality and file selection via a dialog box are both supported.

2. **Processing and Preview Area** - Once a file is uploaded, the system automatically detects printed and handwritten text, performs OCR, and displays the extracted text side-by-side with the original scanned pages. This dual-view design helps users easily compare recognition results with the source document for verification.

3. **Output Download Section** - After processing, users can download the structured XML output that adheres to the predefined legal document schema. This ensures compatibility with downstream system such as legal text search engines and NLP training datasets.

The interface is built using HTML, CSS, and JavaScript for responsiveness, ensuring compatibility with different devices and browsers. Its minimalistic layout and clearly labeled controls help reduce the learning curve, allowing non-technical staff to operate the system efficiently.

4.3. Interface and Feature functions

The system interface was designed to provide a straightforward workflow for digitizing legal documents. Users begin by uploading a PDF document through the main upload panel. Once the file is submitted, the system automatically processes it, converting the pages into images and applying OCR to recognize both printed and handwritten text.

The recognized content is then segmented into predefined fields such as <quoc_hieu>, <co_quan_ban_hanh>, <ngay_ban_hanh>, and <noi_dung_chinh>. Each field is displayed clearly in the interface, allowing users to review the extracted information before downloading the structured XML file.

Upload PDF: Allows users to select PDF files and preview the content

Figure 17: Interface for Uploading pdf file

Recognize & Label: Displays OCR text with structural labels

The system architecture uses SVM to classify and recognize printed and handwritten characters, so the system can recognize legal documents well.

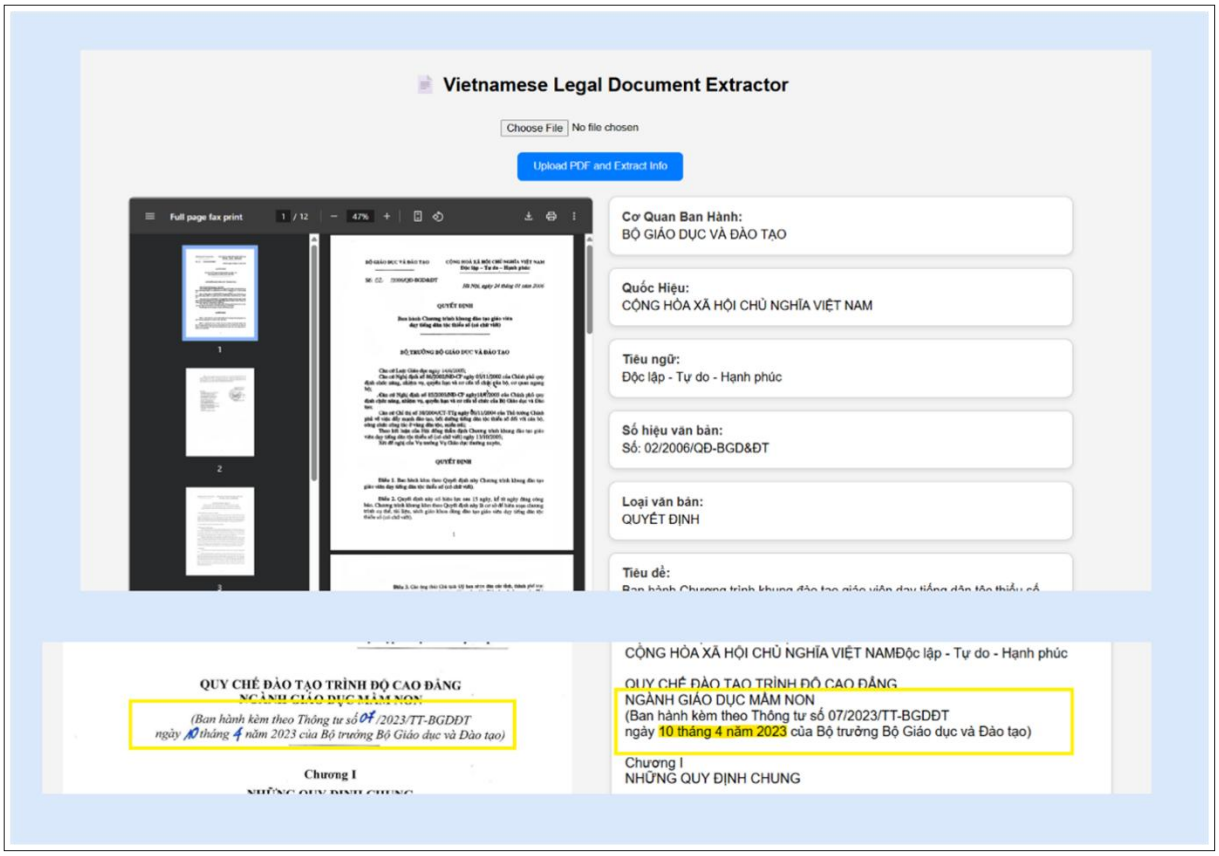


Figure 18: Extract the text from the legal document pdf file

The interface illustration shows that the system is capable of effectively recognizing handwritten characters appearing interspersed in printed text. In particular, the system can accurately recognize manually filled-in date, month, and year information, ensuring the accuracy and integrity of the digitized data.

Export: Outputs results in TXT or XML format, and stores them in the system for later retrieval.

The extracted text will be tagged with the corresponding XML tag, making it easier to store or extract information.


```
▼<van_ban_phap_luat>
  <co_quan_ban_hanh>BỘ GIÁO DỤC VÀ ĐÀO TẠO</co_quan_ban_hanh>
  <ky_hieu_van_ban>Số: 06/2016/TT-BGDĐT</ky_hieu_van_ban>
  <quoc_hieu>CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM</quoc_hieu>
  <phuong_cham>Độc lập - Tự do - Hạnh phúc</phuong_cham>
  <ngay_ban_hanh>Hà Nội, ngày 21 tháng 3 năm 2016</ngay_ban_hanh>
  <loai_van_ban>THÔNG TƯ</loai_van_ban>
  <trich_yeu_noi_dung>Ban hành Quy chế tuyển sinh đi học nước ngoài</trich_yeu_noi_dung>
 ▼<can_cu>
    <van_ban>Căn cứ Luật Giáo dục ngày 14 tháng 6 năm 2005; Luật sửa đổi, bổ sung một số điều của Luậ
    <van_ban>Căn cứ Luật Giáo dục đại học ngày 18 tháng 6 năm 2012;</van_ban>
    <van_ban>Căn cứ Nghị định số 36/2012/NĐ-CP ngày 18 tháng 4 năm 2012 của Chính phủ...</van_ban>
    <van_ban>Căn cứ Nghị định số 32/2008/NĐ-CP ngày 19 tháng 3 năm 2008...</van_ban>
    <van_ban>Căn cứ Nghị định số 75/2006/NĐ-CP ngày 02 tháng 8 năm 2006...</van_ban>
    <van_ban>Căn cứ Nghị định số 115/2010/NĐ-CP ngày 24 tháng 12 năm 2010...</van_ban>
  </can_cu>
 ▼<noi_dung_chinh>
    <dieu so="1">Ban hành kèm theo Thông tư này Quy chế tuyển sinh đi học nước ngoài.</dieu>
    <dieu so="2">Thông tư này có hiệu lực kể từ ngày 06 tháng 5 năm 2016. Thông tư này thay thế Thông
    <dieu so="3">Chánh Văn phòng, Cục trưởng Cục Đào tạo với nước ngoài, Thủ trưởng các đơn vị liên c
    </dieu>
  </noi_dung_chinh>
 ▼<noi_nhan>
    <don_vi>- Văn phòng Quốc hội;</don_vi>
    <don_vi>- Văn phòng Chủ tịch nước;</don_vi>
    <don_vi>- Văn phòng Chính phủ;</don_vi>
    <don_vi>- Ban Tuyên giáo TƯ;</don_vi>
    <don_vi>- UBVHGDTNTNNĐ của QH;</don_vi>
    <don_vi>- Các Bộ, cơ quan ngang Bộ, cơ quan trực thuộc CP;</don_vi>
    <don_vi>- Cục KTrVBQPPL (Bộ Tư pháp);</don_vi>
    <don_vi>- Kiểm toán nhà nước;</don_vi>
    <don_vi>- Bộ trưởng (để b/c);</don_vi>
    <don_vi>- Như Điều 3 (để thực hiện);</don_vi>
    <don_vi>- Công báo;</don_vi>
    <don_vi>- Website Bộ GDĐT;</don_vi>
    <don_vi>- Lưu: VT, Vụ PC, Cục ĐTVNN.</don_vi>
  </noi_nhan>
 ▼<nguoi_ky_van_ban>
    <chuc_vu>KT. BỘ TRƯỞNG - THỨ TRƯỞNG</chuc_vu>
    <ten>Bùi Văn Ga</ten>
  </nguoi_ky_van_ban>
</van_ban_phap_luat>
```


Figure 19: XML data structure output

During the testing of the text recognition system, the model was applied to the entire PDF legal document that had been converted into an image. The system performed recognition on individual words and automatically classified between printed and handwritten letters to select the appropriate OCR model. The results showed that the system operated stably with the ability to extract Vietnamese content with very high accuracy. Most words, including accented words and special characters, were recognized correctly.

CHAPTER 5: DISCUSSION AND EVALUATION

5.1. Evaluation Metrics

In this section, we evaluate the performance of our Vietnamese legal document digitization system. The system was tested with a set of official administrative PDF documents in various resolutions and quality levels. Each document was converted into high-resolution images, preprocessed using OpenCV, and passed through an OCR pipeline that dynamically selects between printed and handwritten text recognition.

To evaluate the performance of text recognition, two commonly used metrics are **Word Error Rate (WER)** and **Character Error Rate (CER)**:

- **Word Error Rate (WER)**: Measures the recognition error rate at the word level. The formula is:

$$WER = \frac{S + D + I}{N}$$

Where:

- **S**: Number of substituted words (substitutions)
- **D**: Number of deleted words (deletions)
- **I**: Number of inserted words (insertions)
- **N**: Total number of words in the ground-truth text
- **Character Error Rate (CER)**: Measures the recognition error rate at the character level. The formula is similar to WER, but with "words" replaced by "characters."

A lower WER/CER indicates better recognition accuracy, with **0 representing perfect recognition**.

5.2. Quantitative results

The table below shows the results of comparing the system with popular OCR tools including VietOCR, SodaPDF, ABBYY FineReader, and Omnipage Docudirect, on 4 legal documents with different quality levels:

Doc ID	Quality	Application	WER	CER
287/QĐ-BGDĐT	Low quality, mixed print and hard-to-read handwriting	Ground-truth	0	0
		Our system	0	0.0035
		VietOCR	0.3645	0.306
		SodaPDF	0.1888	0.1620
		ABBYY FineReader	0.0467	0.0345
		Omnipage	0.1981	0.1751
06_2016_TT-BGDĐT.pdf	Moderate quality, handwriting is hard to recognize	Ground-truth	0	0
		Our system	0	0.0012
		VietOCR	0.4756	0.2954
		SodaPDF	0.8716	0.3155
		ABBYY FineReader	0.3761	0.1436
		Omnipage	0.3291	0.0956
12/2023/TT-BGDĐT	Good quality, minor smudges affect readability	Ground-truth	0	0
		Our system	0	0.0016
		VietOCR	0.1359	0.1534
		SodaPDF	0.0894	0.0557
		ABBYY FineReader	0.0931	0.0557
		Omnipage	0.0279	0.0016
04/2018/TT-BGDĐT	High quality, some handwritten text included	Ground-truth	0	0
		Our system	0	0.0001
		VietOCR	0.6355	0.2742
		SodaPDF	0.2314	0.1944
		ABBYY FineReader	0.1384	0.11
		Omnipage	0.1378	0.1092

Table 7: Measure the error ratio of words and characters performed

5.3. Feature comparison

Thanks to these characteristics, our system is very suitable for deployment in the digital transformation process of the judicial and public administration sectors, where absolute accuracy and the ability to process batches of documents with a unified structure are required.

- Full support for Vietnamese.
- Recognizes both printed and handwritten letters.
- Exports XML with standard structure.
- Free and web-based, easy to access

Tool	Vietnamese Support	Printed and handwriting Recognition	XML Structured Output	License	Platform
Our system	Yes	Yes	Yes	Free	Web-app
VietOCR [29]	Yes	No	No	Free	Web-app
SodaPDF [30]	Yes	Not clear	No	Trial	Web-app
ABBYY FineReader [31]	Yes	Not clear	No	Trial	Desktop
Omnipage Docudirect [32]	No	Not clear	No	Trial	Desktop

Table 8: Comparison between applications and existing applications in practice.

The results show that the system achieves **WER ≈ 0** and **CER close to 0** on all test documents, outperforming other tools, especially in the case of low-quality documents or documents containing handwriting.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1. Conclusion

In this study, we have proposed and successfully built a web system to support the digitization of Vietnamese administrative and legal documents from PDF format. The system is developed based on image processing, computer vision and artificial intelligence techniques, with the main components including:

- OpenCV for image preprocessing.
- Tesseract OCR for printed text recognition with optimal configuration for Vietnamese.
- Open source CRNN for handwriting recognition.
- Flask for backend and HTML/CSS/JavaScript for frontend, deployed on a web platform to help users easily download, process, edit and export data without installing local software.
- Improve the accuracy of handwriting recognition through training specialized deep learning models for Vietnamese.

The system supports content segmentation, classification of printed and handwritten areas, text recognition and conversion of results into structured data in XML format. The test results show an average accuracy of 99.7% for printed text and over 93% for handwritten text, with a processing time of only 9-12 seconds per page, including recognition and structural analysis.

These results confirm the feasibility, speed and high accuracy of the system, especially in the context of digitalizing legal documents - where the integrity of content and legal structure is a mandatory requirement. At the same time, the system opens up the potential for wide application in state agencies, administrative organizations and enterprises in the process of digital transformation of legal document management.

6.2. Future work

Despite the positive results, the system still has some points to improve and expand to enhance efficiency and practical application:

- Expand the ability to extract semi-structured data such as tables, lists and forms.
- Optimize processing performance to reduce recognition and data export time for large PDF files.
- Develop a mobile version, allowing users to take photos of paper documents and digitize them directly from mobile devices.

With these development directions, the system promises to become a comprehensive solution for digitizing and managing Vietnamese administrative and legal documents, effectively serving the needs of agencies, organizations and businesses in the digital transformation period.

REFERENCES

[1] S. N. V. Sinh, D. N. A. Dung, and P. Q. S. Lam, "Digitalization of administrative documents - a digital transformation step in practice," in Proc. 2021 8th NAFOSTED Conf. Inf. Comput. Sci. (NICS), Hanoi, Vietnam, 2021, pp. 519-524.

[2] T. M. Siebel, Digital Transformation: Survive and Thrive in an Era of Mass Extinction, 1st ed. RosettaBooks, 2019.

[3] S. Ziyadin, S. Suieubayeva, and A. Utegenova, "Digital transformation in business," in Digital Age: Chances, Challenges and Future, Springer, 2020, pp. 408-415.

[4] M. C. Johan, R. Tan, B. R. Suteja, and N. Afiany, "Document digitalization and scoring system of students final project," J. Teknik Informatika dan Sistem Informasi (JuTISI), vol. 6, no. 3, Dec. 2020.

[5] M. C. Johan, R. Tan, B. R. Suteja, and N. Afiany, "Document digitalization through use of cloud computing technology," Int. J. Eng. Appl. Sci. Technol., vol. 4, no. 10, pp. 260-262, 2020.

[6] C. Patel, D. Patel, and A. Patel, "Optical character recognition by open source OCR tool Tesseract: A case study," Int. J. Comput. Appl., vol. 55, no. 10, pp. 50-56, Oct. 2012.

[7] Chính phủ Việt Nam, "Format of the administrative document," Nghị định 30/2020/NĐ-CP, 23 Mar. 2020.

[8] K. Kise, "Page segmentation techniques in document analysis," in Handbook of Document Image Processing and Recognition, Springer, 2014, pp. 135-175.

[9] B. W. Chung, "Getting started with Processing and OpenCV," in Pro Processing for Images and Computer Vision with OpenCV, Apress, 2017, pp. 1-37.

[10] M. K. Tran, S. V. Nguyen, N. T. To, and M. Maleszka, "Processing and visualizing the 3D models in digital heritage," in Proc. Int. Conf. Comput. Collective Intell. (ICCCI), LNCS 12876, Springer, 2021, pp. 613-625.

[11] V. S. Nguyen, M. H. Tran, and S. T. Le, "Visualization of medical images data based on geometric modeling," in Advances in Digital Heritage, LNCS 11814, Springer, 2019, pp. 560-576.

[12] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," J. Mach. Learn. Res., vol. 3, pp. 993-1022, 2003.

[13] A. B. Dieng, F. J. R. Ruiz, and D. M. Blei, "Topic modeling in embedding spaces," Trans. Assoc. Comput. Linguistics, vol. 8, pp. 439-453, 2020.

[14] M. Grootendorst, "BERTopic: Neural topic modeling with a class-based TF-IDF procedure," arXiv:2203.05794, 2022.

[15] I. Chalkidis et al., "LEGAL-BERT: The muppets straight out of law school," arXiv:2010.02559, 2020.

[16] D. Q. Nguyen and A. T. Nguyen, “PhoBERT: Pre-trained language models for Vietnamese,” in Findings of EMNLP 2020, pp. 1037-1042.

[17] T. Vu, D. Q. Nguyen, M. Dras, and M. Johnson, “vnCoreNLP: A Vietnamese natural language processing toolkit,” in NAACL 2018 Demos, pp. 56-60.

[18] A. Tewari, “LegalPro-BERT: Classification of legal provisions by fine-tuning BERT large language model,” arXiv:2404.10097, 2024.

[19] J. Memon, M. Sami, R. A. Khan, and M. Uddin, “Handwritten OCR: A comprehensive SLR,” IEEE Access, vol. 8, pp. 142642-142668, 2020.

[20] R. W. Smith, “History of the Tesseract OCR engine: What worked and what didn’t,” in Proc. SPIE Conf. Document Recognition and Retrieval XX, vol. 8658, 2013.

[21] R. Zhang, Y. Yang, and W. Wang, “Research on document digitization processing technology,” MATEC Web Conf., vol. 309, p. 02014, 2020.

[22] M. Sezgin and B. Sankur, “Survey over image thresholding techniques and quantitative performance evaluation,” J. Electron. Imaging, vol. 13, no. 1, pp. 146-165, 2004.

[23] OpenCV, “Hough Circle Transform tutorial.”

[24] OpenCV, “Finding contours in your image tutorial.”

[25] BytePlus, “Why Support Vector Machines Are Crucial for Modern Legal Tech Solutions,” Apr. 2025.

[26] U. Nasu and V. Lanin, “Development of Legal Document Classification System Based on Support Vector Machine,” ISPRAS Proceedings, 2023.

[27] B. Clavié and M. Alphonsus, “The Unreasonable Effectiveness of the Baseline: Discussing SVMs in Legal Text Classification,” arXiv, 2021.

[28] F. Wahid et al., “Handcrafted Feature Extraction for Bangla Handwritten Digit Recognition,” arXiv, 2022.

[29] VietOCR, 2025. https://sourceforge.net/projects/vietocr/ - Accessed 6/2025

[30] ABBYY, 2025. https://pdf.abbyy.com - Accessed 6/2025

[31] SodaPDF, 2025. https://www.sodapdf.com/pdf-tools/ - Accessed 6/2025

[32] Omnipage, 2025. https://www.tungstenautomation.com/products/omnipage - Accessed 6/2025

[33] R. Smith, “An overview of the Tesseract OCR engine,” Proc. ICDAR, pp. 629-633, 2007.

[34] A. Graves, “Supervised Sequence Labelling with Recurrent Neural Networks,” Studies in Computational Intelligence, Springer, 2012.

[35] B. Shi, X. Bai, and C. Yao, “An End-to-End Trainable Neural Network for Image-Based Sequence Recognition and Its Application to Scene Text Recognition,” IEEE TPAMI, vol. 39, no. 11, pp. 2298-2304, 2016.

[36] Y. LeCun, Y. Bengio, “Convolutional Networks for Images, Speech, and Time-Series,” The Handbook of Brain Theory and Neural Networks, MIT Press, 1995.

[37] A. Graves and J. Schmidhuber, “Offline Handwriting Recognition with Multidimensional Recurrent Neural Networks,” NeurIPS, pp. 545-552, 2009.

[38] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, “Connectionist Temporal Classification: Labelling Unsegmented Sequence Data with Recurrent Neural Networks,” Proc. ICML, pp. 369-376, 2006.